

stat_ddd_overid_efficiency

Specialization: v1

TL;DR. The causal aggregate $\tau_\omega(Q)$ is identified by $\Theta_\omega^{\text{agg}}(P)$. Conditional on supplied cross-fitted nuisance learners satisfying the displayed $L_4(P_X)$ and product rates, an everywhere-defined clipped, projected, and ridged estimator has a pointwise Gaussian limit, consistent variance estimate, and Wald coverage. The incremental contribution is the staggered-DDD shared-cell score/covariance mapping, joint constant-span weighting for the prespecified scalar aggregate, the exact componentwise-versus-joint equality frontier and quadratic gap, a feasible descriptive loss diagnostic, and a bounded open nonzero-effect strict-gap family. Shared-unit covariance accounting and generic constrained-GMM/SUR quadratic algebra are prior art. Caron’s subgroup-ATT semiparametric efficiency result concerns a distinct target; no full efficient influence function, semiparametric efficiency bound, uniform learner availability or coverage, or covariate-varying relative-precision result is claimed here.

1 Project justification

Gap. Ortiz-Villavicencio and Sant’Anna optimize admissible comparison cohorts separately within each $ATT(g, t)$ block before prespecified event-study aggregation. Hsieh proves fixed-weight stacked-DDD inference and correct original-unit covariance aggregation when comparison units recur. Neither comparator jointly optimizes the complete retained pairwise-score vector for a prespecified scalar staggered-DDD aggregate or supplies its exact componentwise-versus-joint equality frontier. Caron derives a semiparametric bound for a different causal target—the difference in subgroup ATTs under heterogeneous treatment effects—so it does not answer the retained-score-span weighting question here.

Niche. Repeated cohort-eligibility cells induce cross-block covariance in the retained covariate-adjusted pairwise DDD scores. The DDD-specific shared-cell map makes that covariance computable and shows when zero-sum comparison contrasts from one group-event block can reduce the aggregate variance without altering the prespecified target weights.

Fill. The note identifies the causal aggregate, derives the clipped cross-fitted score expansion and full score-covariance consistency, constructs the feasible joint constant-span estimator and pointwise Wald interval, and proves the exact equality criterion and variance-loss quadratic. It also supplies a consistent descriptive gap diagnostic, an evaluable labelled-invariant restatement, and a bounded open nonzero-effect strict-gap family. The finite-dimensional constrained minimization is used as prior GMM/SUR algebra rather than claimed as a new general efficiency theorem or as a substitute for Caron’s distinct-target semiparametric bound.

2 Related work

Ortiz-Villavicencio and Sant’Anna (2025, Theorem 4.1, Corollary 4.1, and Remark 4.6) supply the retained pairwise staggered-DDD scores and optimize comparison cohorts separately within each $ATT(g, t)$ block before aggregation. Hsieh (2026, Theorem 6.3 and Propositions 6.4–6.5) supplies fixed-weight stacked-DDD asymptotic normality and correct original-unit covariance aggregation when comparison units recur; shared-unit covariance accounting is therefore not claimed as new. Neither result jointly optimizes the complete retained score vector for the prespecified scalar aggregate or states the exact componentwise-versus-joint equality frontier. Caron (2025) derives semiparametric efficiency bounds and doubly robust efficient estimators for the causal difference in subgroup average treatment effects under heterogeneous treatment effects. Its target differs from this note’s prespecified aggregate of eligible-cohort effects in a staggered panel, and its semiparametric bound does not subsume the deliberately narrower question of joint weighting within the fixed retained pairwise-score span. Hansen (1982), Zellner (1962), and Akerberg, Chen, Hahn, and Liao (2014) supply generic finite-dimensional GMM, SUR, and two-step semiparametric geometry, which are prior algebra here. Chen, Sant’Anna, and Xie (2025) inform conditional covariance weighting for staggered two-way difference-in-differences, and Akbari, Kiyavash, and Ghassami (2025) treat canonical two-period DDD efficiency. The scalar central limit theorem used in the proofs is the $k_n = n$ specialization of van der Vaart (1998, Proposition 2.27).

3 Setup

Target (estimand / structural parameter): $\tau_\omega(Q) = \sum_{k \in \mathcal{K}} \omega_k E_Q[Y_{it}(g, 1) - Y_{it}(\infty, 1) \mid G_i = g, S_i = 1]$. Primary functional / estimator / diagnostic:

$$\tau_\omega(Q) = \Theta_\omega^{\text{agg}}(P) = \sum_{k \in \mathcal{K}} \omega_k |\mathcal{J}_k|^{-1} \sum_{c \in \mathcal{J}_k} E_P[p_{g,1}(X_i) d_{k,c}(X_i)] / \pi_{g,1}, \quad P = Q_O$$

- `n` — positive integer; $\mathbb{N}$

Definition. The number of sampled units.

- `T` — fixed positive integer; $\{3, 4, \dots\}$

Definition. The number of balanced-panel periods.

- `time_set` — finite index set; $2^{\mathbb{N}}$

Definition. $\mathcal{T} = \{1, \dots, T\}$.

- `adoption_set` — finite treated-cohort index set; $2^{\mathcal{T}}$

Definition. $\mathcal{G} = \{2, \dots, T\}$ is the set of admissible enabling dates, so every treated cohort has an observed baseline period $g - 1 \in \mathcal{T}$.

- `eligibility_index` — generic eligibility value; $\{0, 1\}$

Definition. s indexes treatment eligibility.

- `cohort_index` — generic enabling-cohort value; $\mathcal{G} \cup \{\infty\}$

Definition. a indexes an enabling cohort.

- `treated_cohort` — generic treated-cohort value; $\mathcal{G} = \{2, \dots, T\}$

Definition. $g \in \mathcal{G}$ indexes a cohort whose effect is targeted and always has the observed baseline $g - 1 \in \mathcal{T}$.

- `post_period` — generic period value; $\mathcal{T}$

Definition. t indexes a post-enabling outcome period.

- `comparison_cohort` — generic comparison-cohort value; $\mathcal{G} \cup \{\infty\}$

Definition. c indexes a cohort not yet enabled at the target period.

- `X_space` — standard Borel space

Definition. $\mathcal{X}$ is the support of pretreatment covariates.

- `covariate_value` — generic covariate value; $\mathcal{X}$

Definition. $x \in \mathcal{X}$.

- `X_i` — random covariate; $\mathcal{X}$

Definition. X_i is unit i 's observed pretreatment covariate.

- `G_i` — random enabling date; $\mathcal{G} \cup \{\infty\}$

Definition. G_i is unit i 's first enabling date, with ∞ denoting never enabled.

- `S_i` — random eligibility indicator; $\{0, 1\}$

Definition. S_i is unit i 's time-invariant eligibility status.

- `Y_pot` — potential-outcome array; $\mathbb{R}^{\mathbb{N} \times \mathcal{T} \times (\mathcal{G} \cup \{\infty\}) \times \{0, 1\}}$

Definition. $Y_{it}(a, s)$ is the period- t outcome under enabling date a and eligibility state s .

- **Y_obs** — observed outcome vector; $\mathbb{R}^T$

Definition. The balanced observed outcome history for unit i .

- **W_i** — full-data causal random element; W

Definition. $W_i = (\{Y_{it}(a, s) : t \in \mathcal{T}, a \in \mathcal{G} \cup \{\infty\}, s \in \{0, 1\}\}, Y_i, G_i, S_i, X_i)$ carries every potential outcome, the observed outcome history, enabling date, eligibility, and pretreatment covariates.

- **Q** — full-data causal law; $\mathcal{P}(W)$

Definition. Q is the common full-data law of W_i .

- **observation_map** — measurable observation map; $o : W \rightarrow O$

Definition. $o(W_i) = (Y_i, G_i, S_i, X_i)$ discards the unobserved potential-outcome coordinates.

- **O_i** — observed-data random element; $\mathbb{R}^T \times (\mathcal{G} \cup \{\infty\}) \times \{0, 1\} \times \mathcal{X}$

Definition. $O_i = o(W_i) = (Y_i, G_i, S_i, X_i)$.

- **P** — observed-data law; $\mathcal{P}(O)$

Definition. $P = Q_O = Q \circ o^{-1}$ is the observed-data pushforward of the full-data law under the observation map.

- **cell_probability** — conditional probability function; $L_2(P_X)$

Definition. $p_{a,s}(x) = P(G_i = a, S_i = s \mid X_i = x)$.

- **target_cell_probability** — positive scalar; $(0, 1)$

Definition. $\pi_{g,1} = P(G_i = g, S_i = 1)$.

- **group_event_index** — generic ordered pair; $\{(g, t) : 2 \leq g \leq t \leq T\}$

Definition. $k = (g, t)$ denotes a treated group-period cell with $2 \leq g \leq t \leq T$; hence the baseline $g - 1$ and post period t are both in $\mathcal{T}$.

- **target_set** — finite nonempty index set; $2^{\{(g,t):2 \leq g \leq t \leq T\}}$

Definition. $\emptyset \neq \mathcal{K} \subseteq \{(g, t) : 2 \leq g \leq t \leq T\}$ is the prespecified support of the event-study aggregate.

- **comparison_set** — finite nonempty index set; $2^{\mathcal{G} \cup \{\infty\}}$

Definition. For $k = (g, t) \in \mathcal{K}$, $\emptyset \neq \mathcal{J}_k \subseteq \{c \in \mathcal{G} \cup \{\infty\} : c > t\}$ is the prespecified set of retained not-yet-enabled comparison cohorts.

- **score_dimension** — positive integer; $\mathbb{N}$

Definition. $M = \sum_{k \in \mathcal{K}} |\mathcal{J}_k|$.

- **outcome_change** — random scalar; $\mathbb{R}$

Definition. For $k = (g, t) \in \mathcal{K}$ with $2 \leq g \leq t \leq T$, $\Delta Y_{i,k} = Y_{it} - Y_{i,g-1}$; in the boundary cohort $g = 2$, this is $Y_{it} - Y_{i1}$.

- **outcome_regression** — conditional mean function; $L_2(P_X)$

Definition. $m_{a,s,k}(x) = E_P[\Delta Y_{i,k} \mid G_i = a, S_i = s, X_i = x]$.

- **ddd_contrast** — conditional DDD contrast function; $L_2(P_X)$

Definition. For $k = (g, t) \in \mathcal{K}$, where $2 \leq g \leq t \leq T$, and $c \in \mathcal{J}_k$, $d_{k,c}(x) = m_{g,1,k}(x) - m_{g,0,k}(x) - m_{c,1,k}(x) + m_{c,0,k}(x)$.

- **residual_covariance** — conditional covariance function; $L_1(P_X)$

Definition.

$$v_{a,s;k\ell}(x) = E_P[(\Delta Y_{i,k} - m_{a,s,k}(x))(\Delta Y_{i,\ell} - m_{a,s,\ell}(x)) \mid G_i = a, S_i = s, X_i = x]$$

- **nuisance_vector** — finite vector of functions; $L_2(P_X)^d$

Definition. η_0 stacks all $p_{a,s}$ and $m_{a,s,k}$ entering the retained comparisons.

- **ATT_vector** — finite-dimensional causal parameter; $\mathbb{R}^{|\mathcal{K}|}$

Definition. For $k = (g, t) \in \mathcal{K}$, where $2 \leq g \leq t \leq T$, $\theta_k(Q) = E_Q[Y_{it}(g, 1) - Y_{it}(\infty, 1) \mid G_i = g, S_i = 1]$.

- **observable_ATT_vector** — finite-dimensional observed-data functional; $\mathbb{R}^{|\mathcal{K}|}$

Definition. For $k = (g, t) \in \mathcal{K}$, where $2 \leq g \leq t \leq T$,

$$\Theta_k(P) = \frac{1}{|\mathcal{J}_k|} \sum_{c \in \mathcal{J}_k} \frac{E_P[p_{g,1}(X_i)d_{k,c}(X_i)]}{\pi_{g,1}}.$$

- **aggregate_weights** — prespecified deterministic vector; $\{w \in \mathbb{R}^{|\mathcal{K}|} : \mathbf{1}'w = 1\}$

Definition. ω fixes the reported event-study aggregation before observing outcomes.

- **tau** — scalar causal parameter; $\mathbb{R}$

Definition. $\tau_\omega(Q) = \omega' \theta(Q) = \sum_{k \in \mathcal{K}} \omega_k \theta_k(Q)$.

- **observable_aggregate** — scalar observed-data functional; $\mathbb{R}$

Definition. $\Theta_\omega^{\text{agg}}(P) = \omega' \Theta(P) = \sum_{k \in \mathcal{K}} \omega_k \Theta_k(P)$.

- **cell_sign** — deterministic sign map; $\{-1, 0, 1\}$

Definition. $\sigma_{g,1}^{g,c} = 1$, $\sigma_{g,0}^{g,c} = -1$, $\sigma_{c,1}^{g,c} = -1$, $\sigma_{c,0}^{g,c} = 1$, and the sign is zero otherwise.

- **score_residual_loading** — deterministic loading function; $L_2(P_X)$

Definition. For $k = (g, t) \in \mathcal{K}$, where $2 \leq g \leq t \leq T$, $c \in \mathcal{J}_k$, and $a \in \{g, c\}$, $L_{k,c}^{a,s}(x) = p_{g,1}(x)\sigma_{a,s}^{g,c}/\{\pi_{g,1}p_{a,s}(x)\}$.

- **pairwise_score** — square-integrable scalar function; $L_2(P)$

Definition. The ratio-adjusted doubly robust pseudo-outcome defined in Def. 3.

- **stacked_score** — random vector; $\mathbb{R}^M$

Definition. $R(O_i; \eta_0) = (R_{k,c}(O_i; \eta_0) : (k, c) \in \mathcal{I})'$, where $\mathcal{I} = \{(k, c) : k \in \mathcal{K}, c \in \mathcal{J}_k\}$.

- **incidence_matrix** — deterministic matrix; $\{0, 1\}^{M \times |\mathcal{K}|}$

Definition. $B_{(k,c),\ell} = \mathbf{1}\{k = \ell\}$, so each row selects the ATT identified by that pairwise score.

- **centered_score** — mean-zero random vector; $L_2^0(P)^M$

Definition. $\Psi(O_i) = R(O_i; \eta_0) - B\Theta(P)$, whose coordinates include the target-share ratio correction in Def. 3.

- **score_covariance** — symmetric matrix; $\mathbb{S}_{++}^M$

Definition. $\Omega = E_P[\Psi(O_i)\Psi(O_i)']$, partitioned into blocks $\Omega_{k\ell}$ conformably with $\mathcal{J}_k$ and $\mathcal{J}_\ell$.

- **ddd_covariance** — symmetric matrix functional; $\mathbb{S}_+^M$

Definition. The primitive shared-cell covariance construction in Def. 4.

- **contrast_matrix** — deterministic block-diagonal matrix; $\mathbb{R}^{M \times (M - |\mathcal{K}|)}$

Definition. Order each $\mathcal{J}_k = \{c_{k,1} < \dots < c_{k,J_k}\}$, with ∞ last when present, and set $N = \text{diag}(N_k : k \in \mathcal{K})$, where $N_k = [e_1 - e_{J_k}, \dots, e_{J_k-1} - e_{J_k}]$. Thus N is a canonical full-column-rank zero-sum contrast basis determined by the labelled comparison sets and satisfies $B'N = 0$.

- **separate_weight** — deterministic score-loading vector; $\mathbb{R}^M$

Definition. The componentwise GMM loading defined in Def. 5.

- **joint_weight** — deterministic score-loading vector; $\mathbb{R}^M$

Definition. The joint aggregate GMM loading defined in Def. 6.

- **separate_variance** — nonnegative scalar; $[0, \infty)$

Definition. $V_{\text{sep}} = (b^{\text{sep}})' \Omega b^{\text{sep}}$.

- **joint_variance** — nonnegative scalar; $[0, \infty)$

Definition. $V_{\text{joint}} = (b^{\text{joint}})' \Omega b^{\text{joint}}$.

- **number_folds** — fixed integer; $\{2, 3, \dots\}$

Definition. The fixed number of cross-fitting folds.

- **fold_map** — deterministic partition map; $\{1, \dots, n\} \rightarrow \{1, \dots, F\}$

Definition. $f(i)$ assigns unit i to one of F folds whose sizes differ by at most one.

- **moment_constant** — fixed positive constant; $(0, \infty)$

Definition. A common fourth-moment bound.

- **spectrum_constant** — fixed constant; $(0, 1)$

Definition. A common covariance-eigenvalue bound.

- **propensity_l4_rate** — deterministic positive sequence; $\mathbb{R}_+^{\mathbb{N}}$

Definition. The fold-uniform $L_4(P_X)$ rate for raw cell-propensity fits, with $a_n \rightarrow 0$.

- **regression_l4_rate** — deterministic positive sequence; $\mathbb{R}_+^{\mathbb{N}}$

Definition. The fold-uniform $L_4(P_X)$ rate for outcome-regression fits, with $b_n \rightarrow 0$ and $a_n b_n = o(n^{-1/2})$.

- **ridge_sequence** — deterministic positive sequence; $(0, \infty)^{\mathbb{N}}$

Definition. $\lambda_n = n^{-1/4}$ is the strictly positive covariance ridge.

- **test_level** — fixed constant; $(0, 1)$

Definition. The two-sided Wald test size.

- **normal_quantile** — positive scalar; $(0, \infty)$

Definition. $z_{1-\alpha/2}$ is the $1 - \alpha/2$ quantile of $N(0, 1)$.

- **estimated_nuisance** — off-fold function estimate; $L_4(P_X)^d$

Definition. $\hat{\eta}_{-f(i)}$ stacks real-valued raw fits $\hat{p}_{a,s,-f(i)}$ and $\hat{m}_{a,s,k,-f(i)}$ trained without fold $f(i)$.

- **score_coordinate_set** — finite labelled index set; $2^{\mathcal{K} \times (\mathcal{G} \cup \{\infty\})}$

Definition. $\mathcal{I} = \{j = (k, c) : k \in \mathcal{K}, c \in \mathcal{J}_k\}$ is the labelled set of retained pairwise-score coordinates.

- **primitive_residual_set** — finite labelled residual-cell index set; $2^{(\mathcal{G} \cup \{\infty\}) \times \{0,1\} \times \mathcal{T}}$

Definition.

$\mathcal{R} = \{\rho = (a, s, u) : \text{for some } j = ((g, t), c) \in \mathcal{I}, a \in \{g, c\}, s \in \{0, 1\}, u \in \{g - 1, t\}\}$
 records every observed cohort, eligibility, and time coordinate underlying a retained outcome-change residual.

- **signed_incidence_tensor** — labelled measurable incidence tensor; $L_2(P_X)^{\mathcal{I} \times \mathcal{R}}$

Definition. For $j = ((g, t), c) \in \mathcal{I}$ and $\rho = (a, s, u) \in \mathcal{R}$, set

$$A_{j\rho}(x) = L_{(g,t),c}^{a,s}(x) \mathbf{1}\{a \in \{g, c\}\} \{\mathbf{1}(u = t) - \mathbf{1}(u = g - 1)\}.$$

This records the score label, admissible comparison label, cohort-eligibility-time residual cell, and its signed coefficient.

- **compatible_covariance_map** — labelled covariance-map tensor; $\mathbb{S}_+^M$

Definition. For $j = (k, c)$ and $q = (\ell, d)$, $C_{jq}(P)$ is the right-hand side of the primitive shared-cell formula in Def. 4, including both the target-cell contrast covariance and every residual covariance entry induced by the signed incidences $A_{j\rho}$ and $A_{q\rho'}$. Thus $\mathfrak{C}(P) = \Omega(P)$ for compatible observed-data laws.

- **labelled_reuse_invariant** — law-indexed labelled multigraph and incidence invariant; $\mathfrak{H}(P)$

Definition. $\mathfrak{H}(P) = (\mathcal{I}, (\mathcal{J}_k)_{k \in \mathcal{K}}, \mathcal{R}, A, \omega, \mathfrak{C}(P), N)$ retains every score coordinate $j = (k, c)$, every admissible comparison label, every cohort-eligibility-time residual coordinate, every signed incidence coefficient, the prespecified aggregate weights, all compatible covariance-map entries needed to recover $\Omega(P)$, and the canonical contrast matrix. Two invariants are equivalent only when there are label-preserving bijections under which all seven components, including A , ω , $\mathfrak{C}$, and N , agree exactly.

- **retained_cell_count** — fixed positive integer; $\mathbb{N}$

Definition. Define the fixed retained cohort-eligibility cell set by

$$\mathcal{C}_{\text{ret}} = \{(a, s) : s \in \{0, 1\}, \exists k = (g, t) \in \mathcal{K}, c \in \mathcal{J}_k, a \in \{g, c\}\}$$

, and set $L_{\text{ret}} = |\mathcal{C}_{\text{ret}}|$.

- **overlap_constant** — fixed constant; $\{u \in \mathbb{R} : 0 < u < L_{\text{ret}}^{-1}\}$

Definition. A feasible common lower bound for all distinct retained cohort-eligibility cells in the fixed design.

- **clipping_map** — deterministic scalar map; $\mathbb{R} \rightarrow [\varepsilon/2, 1 - \varepsilon/2]$

Definition. $C_\varepsilon(u) = \min\{1 - \varepsilon/2, \max\{\varepsilon/2, u\}\}$.

- **clipped_propensity** — off-fold bounded function estimate; $L_2(P_X)$

Definition. $\tilde{p}_{a,s,-f(i)}(x) = C_\varepsilon(\hat{p}_{a,s,-f(i)}(x))$.

- **clipped_target_probability** — random bounded scalar; $[\varepsilon/2, 1 - \varepsilon/2]$

Definition. $\tilde{\pi}_{g,1} = C_\varepsilon(n^{-1} \sum_{i=1}^n \mathbf{1}\{G_i = g, S_i = 1\})$.

- **fitted_influence** — finite random vector; $\mathbb{R}^M$

Definition. The everywhere-defined cross-fitted influence vector displayed in Def. 7.

- **score_mean_estimator** — random vector; $\mathbb{R}^M$

Definition. The vector of cross-fitted Hájek-ratio pairwise DDD estimators defined in Def. 7.

- **raw_covariance_estimator** — random symmetric matrix; $\mathbb{S}^M$

Definition. $\hat{\Omega}^{\text{raw}} = n^{-1} \sum_{i=1}^n \hat{\Psi}_i \hat{\Psi}_i'$.

- `covariance_estimator` — random symmetric matrix; $\mathbb{S}_+^M$

Definition. $\hat{\Omega} = \Pi_{\mathbb{S}_+^M}(\hat{\Omega}^{\text{raw}})$, where $\Pi_{\mathbb{S}_+^M}$ is the unique Frobenius-norm projection onto the positive-semidefinite cone.

- `regularized_covariance_estimator` — random positive-definite matrix; $\mathbb{S}_{++}^M$

Definition. $\hat{\Omega}_\lambda = \hat{\Omega} + \lambda_n I_M$, which is positive definite for every realized sample.

- `estimated_separate_weight` — random score-loading vector; $\mathbb{R}^M$

Definition. For each k ,

$$\hat{b}_k^{\text{sep}} = \omega_k \hat{\Omega}_{\lambda, kk}^{-1} \mathbf{1} / (\mathbf{1}' \hat{\Omega}_{\lambda, kk}^{-1} \mathbf{1})$$

, and the blocks are stacked.

- `estimated_joint_weight` — random score-loading vector; $\mathbb{R}^M$

Definition.

$$\hat{b}^{\text{joint}} = \hat{\Omega}_\lambda^{-1} B (B' \hat{\Omega}_\lambda^{-1} B)^{-1} \omega$$

- `joint_estimator` — random scalar; $\mathbb{R}$

Definition. The feasible joint aggregate estimator defined in Def. 7.

- `separate_estimator` — random scalar; $\mathbb{R}$

Definition. The feasible componentwise-GMM plug-in aggregate defined in Def. 7.

- `joint_variance_estimator` — random nonnegative scalar; $[0, \infty)$

Definition.

$$\hat{V}_{\text{joint}} = (\hat{b}^{\text{joint}})' \hat{\Omega} \hat{b}^{\text{joint}}$$

, using the projected but unridged covariance.

- `wald_interval` — random closed interval; $2^{\mathbb{R}}$

Definition.

$$\mathcal{C}_{n, 1-\alpha} = [\hat{\tau}_{\text{joint}} - z_{1-\alpha/2} \sqrt{\hat{V}_{\text{joint}}/n}, \hat{\tau}_{\text{joint}} + z_{1-\alpha/2} \sqrt{\hat{V}_{\text{joint}}/n}]$$

; when $\hat{V}_{\text{joint}} = 0$, its standard error is defined as zero and the interval is the singleton $\{\hat{\tau}_{\text{joint}}\}$.

- `estimated_variance_gap` — random nonnegative scalar; $[0, \infty)$

Definition.

$$\hat{\Delta} = (\hat{b}^{\text{sep}} - \hat{b}^{\text{joint}})' \hat{\Omega} (\hat{b}^{\text{sep}} - \hat{b}^{\text{joint}})$$

is the feasible diagnostic for the constant-score-span variance loss.

4 Assumptions

Assumption 1 (ass:iid-panel). $W_1, \dots, W_n$ are independent draws from Q , with fixed T .

Standard (Independent balanced-panel sampling, [1]).

Assumption 2 (ass:consistency). $Q\{Y_{it} = Y_{it}(G_i, S_i) \text{ for every } t \in \mathcal{T}\} = 1$.

Standard (Potential-outcome consistency, [9]).

Assumption 3 (ass:no-anticipation).

$Q\{Y_{it}(g, 1) = Y_{it}(\infty, 1) \text{ for every } g \in \mathcal{G} \text{ and } t \in \mathcal{T} \text{ with } t < g\} = 1$

Standard (No anticipation under staggered adoption, [1]).

Assumption 4 (ass:no-ineligible-spillover).

$Q\{Y_{it}(a, 0) = Y_{it}(\infty, 0) \text{ for every } a \in \mathcal{G} \cup \{\infty\} \text{ and } t \in \mathcal{T}\} = 1$

Standard (No treatment effect on ineligible units, [1]).

Assumption 5 (ass:pairwise-conditional-ddd). For every $k = (g, t) \in \mathcal{K}$ with $2 \leq g \leq t \leq T$, every $c \in \mathcal{J}_k$, and Q_X -almost every x ,

$$E_Q[Y_{it}(\infty, 1) - Y_{i,g-1}(\infty, 1) \mid G_i = g, S_i = 1, X_i = x] - E_Q[Y_{it}(\infty, 0) - Y_{i,g-1}(\infty, 0) \mid G_i = g, S_i = 0, X_i = x] = E_Q[Y_{it}(\infty, 1) - Y_{i,g-1}(\infty, 1) \mid G_i = c, S_i = 1, X_i = x] - E_Q[Y_{it}(\infty, 0) - Y_{i,g-1}(\infty, 0) \mid G_i = c, S_i = 0, X_i = x]$$

Standard (Pairwise covariate-conditional DDD parallel trends, [1]).

Assumption 6 (ass:cell-overlap). For every $(a, s) \in \mathcal{C}_{\text{ret}}$, $Q(G_i = a, S_i = s \mid X_i = x) \geq \varepsilon$ for Q_X -almost every x , where $0 < \varepsilon < L_{\text{ret}}^{-1}$.

Standard (Uniform generalized-propensity overlap, [1]).

Assumption 7 (ass:fourth-moment). $E_P[\max_{t \in \mathcal{T}} |Y_{it}|^4] \leq C_4$.

Standard (Finite fourth-moment GMM regularity, [7]).

Assumption 8 (ass:score-spectrum). $\kappa I_M \preceq \Omega \preceq \kappa^{-1} I_M$.

Standard (Full-rank optimal-GMM covariance, [7]).

Assumption 9 (ass:foldwise-independence). For every fold f , conditional on the training observations $\{O_i : f(i) \neq f\}$, the off-fold fit $\hat{\eta}_{-f}$ is fixed and is independent of the validation observations $\{O_i : f(i) = f\}$.

Standard (Foldwise sample splitting, [8]).

Assumption 10 (ass:propensity-l4-rate). Conditional on the training samples, $\max_{f,a,s} \|\hat{p}_{a,s,-f} - p_{a,s}\|_{P_X,4} = O_P(a_n)$ with $a_n \rightarrow 0$.

Standard (Cross-fitted propensity fourth-norm rate, [8]).

Assumption 11 (ass:regression-l4-rate). Conditional on the training samples, $\max_{f,a,s,k} \|\hat{m}_{a,s,k,-f} - m_{a,s,k}\|_{P_X,4} = O_P(b_n)$ with $b_n \rightarrow 0$.

Standard (Cross-fitted regression fourth-norm rate, [8]).

Assumption 12 (ass:nuisance-product-rate). The deterministic learner rates satisfy $a_n b_n = o(n^{-1/2})$.

Standard (Cross-fitted orthogonal-score product rate, [8]).

5 Definitions

Definition 1 (def:ddd-law-class). Defined object. *Staggered-DDD law class*

Construction.

$$\mathcal{M} = \{Q \in \mathcal{P}(\mathcal{W}) : Q \text{ satisfies consistency, no anticipation, no ineligible spillover, every retained pairwise conditional DDD restriction, cell overlap with } 0 < \varepsilon < L_{\text{ret}}^{-1}, \text{ and, for } P = Q_O, \text{ the outcome fourth-moment and score-spectrum bounds}\}$$

. The class contains only primitive restrictions on Q and its observed-data pushforward P ; conditional Jensen's inequality and overlap imply the needed true-regression moments. Independent sampling, cross-fitting, learner rates, and fitted-regression tightness are theorem-level conditions or consequences, not members of the law class.

Member properties. `ass:consistency`; `ass:no-anticipation`; `ass:no-ineligible-spillover`; `ass:pairwise-conditional-ddd`; `ass:cell-overlap`; `ass:fourth-moment`; `ass:score-spectrum`.

Definition 2 (def:observable-functional). Defined object. *Observed-data DDD functional*

Construction. For $k = (g, t) \in \mathcal{K}$, where $2 \leq g \leq t \leq T$, define

$$\Theta_k(P) = \frac{1}{|\mathcal{J}_k|} \sum_{c \in \mathcal{J}_k} \frac{E_P[p_{g,1}(X_i)d_{k,c}(X_i)]}{\pi_{g,1}}, \quad \Theta_\omega^{\text{agg}}(P) = \omega' \Theta(P).$$

Both quantities are functionals only of the observed-data law $P = Q_O$.

Inputs. *def:ddd-law-class*.

Definition 3 (def:pairwise-score). Defined object. *Pairwise ratio-adjusted DDD score*

Construction. For $k = (g, t) \in \mathcal{K}$, where $2 \leq g \leq t \leq T$, and $c \in \mathcal{J}_k$, set

$$R_{k,c}(O_i; \eta_0) = \Theta_k(P) + \frac{\mathbf{1}\{G_i = g, S_i = 1\}}{\pi_{g,1}} \{d_{k,c}(X_i) - \Theta_k(P)\} + \frac{p_{g,1}(X_i)}{\pi_{g,1}} \sum_{(a,s) \in \{g,c\} \times \{0,1\}} \frac{\sigma_{a,s}^{g,c} \mathbf{1}\{G_i = a, S_i = s\}}{p_{a,s}(X_i)} \{\Delta Y_{i,k} - m_{a,s,k}(X_i)\}.$$

Thus $R_{k,c} - \Theta_k(P)$ is the observed-data first-order influence of the Hájek-ratio pairwise estimator, including estimation of $\pi_{g,1}$.

Inputs. *def:ddd-law-class; def:observable-functional*.

Definition 4 (def:shared-covariance). Defined object. *Primitive shared-cell covariance matrix*

Construction. For score coordinates $j = (k, c)$ and $q = (\ell, d)$, let g_k and g_ℓ denote the first components of k and ℓ , and define

$$\Omega_{jq}^{\text{DDD}} = E_P \left[\frac{\mathbf{1}\{g_k = g_\ell\} p_{g_k,1}(X_i) \{d_{k,c}(X_i) - \Theta_k(P)\} \{d_{\ell,d}(X_i) - \Theta_\ell(P)\}}{\pi_{g_k,1} \pi_{g_\ell,1}} + \sum_{a,s} p_{a,s}(X_i) L_{k,c}^{a,s}(X_i) L_{\ell,d}^{a,s}(X_i) v_{a,s;k\ell}(X_i) \right].$$

The sum ranges over cohort-eligibility cells entering both score coordinates.

Inputs. *def:pairwise-score*.

Definition 5 (def:separate-loading). Defined object. *Componentwise GMM aggregate loading*

Construction. For every $k \in \mathcal{K}$,

$$(b^{\text{sep}})_k = \omega_k \frac{\Omega_{kk}^{-1} \mathbf{1}}{\mathbf{1}' \Omega_{kk}^{-1} \mathbf{1}},$$

and b^{sep} stacks these blocks. This is exactly the loading produced by optimizing each group-event cell separately and then applying ω .

Inputs. *def:pairwise-score*.

Definition 6 (def:joint-loading). Defined object. *Joint aggregate GMM loading*

Construction.

$$b^{\text{joint}} = \Omega^{-1} B (B' \Omega^{-1} B)^{-1} \omega.$$

Inputs. *def:pairwise-score*.

Definition 7 (def:joint-estimator). Defined object. *Feasible separate and joint estimators*

Construction. For $k = (g, t) \in \mathcal{K}$, where $2 \leq g \leq t \leq T$, $c \in \mathcal{J}_k$, and observation i , first set $\tilde{p}_{a,s,i} = C_\varepsilon(\hat{p}_{a,s,-f(i)}(X_i))$, $\tilde{p}_{g,1,i} = C_\varepsilon(\hat{p}_{g,1,-f(i)}(X_i))$, $\tilde{\pi}_{g,1} = C_\varepsilon(n^{-1} \sum_{j=1}^n \mathbf{1}\{G_j = g, S_j = 1\})$, and

$\hat{d}_{k,c,i} = \hat{m}_{g,1,k,-f(i)}(X_i) - \hat{m}_{g,0,k,-f(i)}(X_i) - \hat{m}_{c,1,k,-f(i)}(X_i) + \hat{m}_{c,0,k,-f(i)}(X_i)$
. Define

$$\hat{A}_{i,k,c} = \mathbf{1}\{G_i = g, S_i = 1\} \hat{d}_{k,c,i} + \tilde{p}_{g,1,i} \sum_{(a,s) \in \{g,c\} \times \{0,1\}} \frac{\sigma_{a,s}^{g,c} \mathbf{1}\{G_i = a, S_i = s\}}{\tilde{p}_{a,s,i}} \{\Delta Y_{i,k} - \hat{m}_{a,s,k,-f(i)}(X_i)\}, \quad \hat{\mu}_{k,c} = \frac{n^{-1} \sum_{i=1}^n \hat{A}_{i,k,c}}{\tilde{\pi}_{g,1}},$$

and display the fitted influence coordinate

$$\hat{\Psi}_{i,k,c} = \frac{1}{\tilde{\pi}_{g,1}} \left[\mathbf{1}\{G_i = g, S_i = 1\} \{\hat{d}_{k,c,i} - \hat{\mu}_{k,c}\} + \tilde{p}_{g,1,i} \sum_{(a,s) \in \{g,c\} \times \{0,1\}} \frac{\sigma_{a,s}^{g,c} \mathbf{1}\{G_i = a, S_i = s\}}{\tilde{p}_{a,s,i}} \{\Delta Y_{i,k} - \hat{m}_{a,s,k,-f(i)}(X_i)\} \right].$$

Stack $\hat{\mu}_{k,c}$ and $\hat{\Psi}_{i,k,c}$. Set $\hat{\Omega}^{\text{raw}} = n^{-1} \sum_i \hat{\Psi}_i \hat{\Psi}_i'$, $\hat{\Omega} = \Pi_{\mathbb{S}_+^M}(\hat{\Omega}^{\text{raw}})$, and $\hat{\Omega}_\lambda = \hat{\Omega} + n^{-1/4} I_M$. For each k , set

$$\hat{b}_k^{\text{sep}} = \omega_k \hat{\Omega}_{\lambda, kk}^{-1} \mathbf{1} / (\mathbf{1}' \hat{\Omega}_{\lambda, kk}^{-1} \mathbf{1})$$

, and set

$$\hat{b}^{\text{joint}} = \hat{\Omega}_\lambda^{-1} B (B' \hat{\Omega}_\lambda^{-1} B)^{-1} \omega$$

. Finally, $\hat{\tau}_{\text{sep}} = (\hat{b}^{\text{sep}})' \hat{\mu}$, $\hat{\tau}_{\text{joint}} = (\hat{b}^{\text{joint}})' \hat{\mu}$, and

$$\hat{V}_{\text{joint}} = (\hat{b}^{\text{joint}})' \hat{\Omega} \hat{b}^{\text{joint}}$$

. Define the feasible constant-score-span efficiency-loss diagnostic by

$$\hat{\Delta} = (\hat{b}^{\text{sep}} - \hat{b}^{\text{joint}})' \hat{\Omega} (\hat{b}^{\text{sep}} - \hat{b}^{\text{joint}}).$$

Every denominator and inverse in this construction is defined for every realized sample, and $\hat{\Delta} \geq 0$ because $\hat{\Omega}$ is positive semidefinite.

Inputs. `def:separate-loading`; `def:joint-loading`.

Definition 8 (`def:wald-interval`). Defined object. *Studentized joint-GMM Wald interval*

Construction. Let $z_{1-\alpha/2}$ be the $1 - \alpha/2$ quantile of $N(0, 1)$ and define

$$\mathcal{C}_{n, 1-\alpha} = \left[\hat{\tau}_{\text{joint}} - z_{1-\alpha/2} \sqrt{\hat{V}_{\text{joint}}/n}, \hat{\tau}_{\text{joint}} + z_{1-\alpha/2} \sqrt{\hat{V}_{\text{joint}}/n} \right].$$

If $\hat{V}_{\text{joint}} = 0$, define the estimated standard error to be zero, do not divide by it, and set $\mathcal{C}_{n, 1-\alpha} = \{\hat{\tau}_{\text{joint}}\}$.

Inputs. `def:joint-estimator`.

Definition 9 (`def:labelled-incidence-handle`). Defined object. *Sufficient labelled shared-cell invariant*

Construction. Expand every score coordinate $j = ((g, t), c) \in \mathcal{I}$ into the cohort-eligibility-time residual coordinates $\rho = (a, s, u) \in \mathcal{R}$, attach the signed functional coefficient $A_{j\rho}(x)$, and retain $\mathfrak{H}(P) = (\mathcal{I}, (\mathcal{J}_k)_k, \mathcal{R}, A, \omega, \mathfrak{C}(P), N)$. Recover $\Omega(P) = \mathfrak{C}(P)$, compute the published componentwise loading blockwise, and evaluate $N' \Omega(P) b^{\text{sep}}(P)$. Classify only label-preserving equivalence classes for which the comparison labels, residual labels, signed incidences, aggregate weights, covariance-map entries, and canonical contrast matrix all agree. For the smallest strict-gap search, specialize to $T = 3$, $\mathcal{K} = \{(2, 2), (2, 3)\}$, $\mathcal{J}_{(2,2)} = \{3, \infty\}$, and $\mathcal{J}_{(2,3)} = \{\infty\}$; parameterize bounded cell residuals and target-cell contrasts, enforce the causal restrictions, compute the resulting 3×3 positive-definite Ω , and determine the variance-gap quadratic form exactly.

Inputs. `def:pairwise-score`; `def:shared-covariance`; `def:separate-loading`; `def:joint-loading`.

6 Main results

Proposition 1 (prop:class-nonempty). *The primitive law class is nonempty for a feasible common overlap constant. Take $T = 3$, $\mathcal{K} = \{(2, 2)\}$, $\mathcal{J}_{(2,2)} = \{3\}$, $\omega_{(2,2)} = 1$, $\varepsilon = 1/8$, $C_4 = 1$, and $\kappa = 1/2$. Let $X_i = x_0$ almost surely; let (G_i, S_i) be uniform on $\{(2, 0), (2, 1), (3, 0), (3, 1)\}$; and independently let U_{i1}, U_{i2}, U_{i3} be independent Rademacher variables. Set $Y_{it}(a, s) = U_{it}/8$ for every $a \in \{2, 3, \infty\}$, $s \in \{0, 1\}$, and $t \in \{1, 2, 3\}$, and set $Y_{it} = Y_{it}(G_i, S_i)$. The induced full-data law Q_0 belongs to $\mathcal{M}$: all four retained disjoint cells have probability $1/4 > \varepsilon$, the causal restrictions hold with zero effects and common trends, $m_{a,s,(2,2)} = 0$, and the sole centered score equals $4\sigma_{a,s}^{2,3}\{U_{i2} - U_{i1}\}/8$ on cell (a, s) . Consequently, $\Omega = 1/2$, so $\kappa I_1 \preceq \Omega \preceq \kappa^{-1} I_1$, while the outcome and true-regression fourth-moment bounds hold.*

Proof. Let Q_0 be the full-data law specified in the proposition and let $P_0 = Q_0 \circ o^{-1}$ be its observed-data pushforward. For the stated design, the retained cohort–eligibility cells are

$$\mathcal{C}_{\text{ret}} = \{(2, 0), (2, 1), (3, 0), (3, 1)\}, \quad L_{\text{ret}} = 4.$$

Because $X_i = x_0$ almost surely and (G_i, S_i) is uniform on this set, for every $(a, s) \in \mathcal{C}_{\text{ret}}$,

$$p_{a,s}(x_0) = Q_0(G_i = a, S_i = s \mid X_i = x_0) = \frac{1}{4} > \frac{1}{8} = \varepsilon.$$

Moreover, $0 < \varepsilon = 1/8 < L_{\text{ret}}^{-1} = 1/4$. Thus the simultaneous retained-cell overlap restriction is feasible and holds $Q_{0,X}$ -almost surely.

Consistency holds by the stipulated equality $Y_{it} = Y_{it}(G_i, S_i)$. Since

$$Y_{it}(a, s) = \frac{U_{it}}{8}$$

for every enabling date a , eligibility state s , and period t , we have, in particular,

$$Y_{it}(g, 1) = \frac{U_{it}}{8} = Y_{it}(\infty, 1)$$

whenever $t < g$; hence no anticipation holds. The same equality for $s = 0$ gives

$$Y_{it}(a, 0) = \frac{U_{it}}{8} = Y_{it}(\infty, 0),$$

so there is no spillover to ineligible units.

There is only one retained group–period index, $k = (2, 2)$, and only one comparison cohort, $c = 3$. Its observed change is

$$\Delta Y_{i,(2,2)} = Y_{i2} - Y_{i1} = \frac{U_{i2} - U_{i1}}{8}.$$

The Rademacher variables are independent of (G_i, S_i, X_i) and have mean zero. Therefore, for each retained cell,

$$m_{a,s,(2,2)}(x_0) = E_{P_0} \left[\frac{U_{i2} - U_{i1}}{8} \mid G_i = a, S_i = s, X_i = x_0 \right] = 0.$$

Consequently $d_{(2,2),3}(x_0) = 0$. The conditional DDD parallel-trends restriction also holds directly: each of its four conditional untreated mean changes equals

$$E_{Q_0} \left[\frac{U_{i2} - U_{i1}}{8} \mid G_i = a, S_i = s, X_i = x_0 \right] = 0,$$

so both eligibility differences in that restriction are zero. In addition, the target causal effect is identically zero, because

$$Y_{i2}(2, 1) - Y_{i2}(\infty, 1) = 0.$$

It follows from the displayed observed functional that $\Theta_{(2,2)}(P_0) = 0$, while $\pi_{2,1} = 1/4$.

We next verify the moment and spectrum restrictions. Since every observed outcome has absolute value $1/8$,

$$E_{P_0} \left[\max_{t \in \{1, 2, 3\}} |Y_{it}|^4 \right] = \frac{1}{8^4} \leq 1 = C_4.$$

The retained true regressions are zero and hence have every finite moment. Substituting $d_{(2,2),3} = 0$, $\Theta_{(2,2)} = 0$, $p_{a,s} = 1/4$, and $\pi_{2,1} = 1/4$ into the pairwise-score definition shows that, on a realized cell $(G_i, S_i) = (a, s)$,

$$\Psi(O_i) = R_{(2,2),3}(O_i; \eta_0) = 4\sigma_{a,s}^{2,3} \frac{U_{i2} - U_{i1}}{8}.$$

This score has mean zero. Since $(\sigma_{a,s}^{2,3})^2 = 1$ on every cell in the support and the two Rademacher variables are independent,

$$\Omega = E_{P_0}[\Psi(O_i)^2] = 16E\left[\left(\frac{U_{i2} - U_{i1}}{8}\right)^2\right] = \frac{16}{64}\{E(U_{i2}^2) + E(U_{i1}^2) - 2E(U_{i2})E(U_{i1})\} = \frac{1}{2}.$$

Here $M = 1$. With $\kappa = 1/2$, the required scalar matrix inequalities are therefore

$$\kappa I_1 = \frac{1}{2} \leq \Omega = \frac{1}{2} \leq 2 = \kappa^{-1} I_1.$$

All defining restrictions of $\mathcal{M}$ have now been verified for Q_0 , so $Q_0 \in \mathcal{M}$. Hence the primitive law class is nonempty. $\square$

Justification. The explicit four-cell construction remains compatible with the corrected overlap domain because $L_{\text{ret}} = 4$ and $1/8 < 1/4$. *Gap.* This is a fixed-design nonemptiness check, not a learner-achievability or novelty theorem. *Consumer.* It confirms that the corrected law class contains a concrete bounded law.

Lemma 1 (lem:causal-identification). *For every full-data law $Q \in \mathcal{M}$, let $P = Q_O = Q \circ o^{-1}$. For every $k = (g, t) \in \mathcal{K}$, where $2 \leq g \leq t \leq T$, and every $c \in \mathcal{J}_k$,*

$$\theta_k(Q) = \frac{E_P[p_{g,1}(X_i)d_{k,c}(X_i)]}{\pi_{g,1}} = E_P[R_{k,c}(O_i; \eta_0)].$$

Consequently, $\theta(Q) = \Theta(P)$, $\tau_\omega(Q) = \Theta_\omega^{\text{agg}}(P)$, $E_P[R(O_i; \eta_0)] = B\Theta(P)$, and $E_P[\Psi(O_i)] = 0$. For the earliest admissible cohort $g = 2$, every displayed change uses the defined baseline $g - 1 = 1$.

Proof. Fix $k = (g, t) \in \mathcal{K}$ and $c \in \mathcal{J}_k$. For $a \in \{g, c\}$ and $s \in \{0, 1\}$, define the untreated conditional change

$$\mu_{a,s,k}^0(x) = E_Q[Y_{it}(\infty, s) - Y_{i,g-1}(\infty, s) \mid G_i = a, S_i = s, X_i = x]$$

and the target-cell conditional effect

$$q_k(x) = E_Q[Y_{it}(g, 1) - Y_{it}(\infty, 1) \mid G_i = g, S_i = 1, X_i = x].$$

Consistency and no anticipation at $g - 1 < g$ give

$$m_{g,1,k}(x) = q_k(x) + \mu_{g,1,k}^0(x).$$

No spillover to ineligible units gives $m_{g,0,k}(x) = \mu_{g,0,k}^0(x)$. Since $c > t \geq g$, both $g - 1 < c$ and $t < c$. Consistency and no anticipation therefore give

$$m_{c,1,k}(x) = \mu_{c,1,k}^0(x), \quad m_{c,0,k}(x) = \mu_{c,0,k}^0(x),$$

where the second equality also follows from no ineligible spillover. Substitution into the observed conditional DDD contrast yields

$$d_{k,c}(x) = q_k(x) + \{\mu_{g,1,k}^0(x) - \mu_{g,0,k}^0(x)\} - \{\mu_{c,1,k}^0(x) - \mu_{c,0,k}^0(x)\}.$$

The pairwise conditional DDD restriction makes the two braced differences equal for Q_X -almost every x . Thus $d_{k,c}(x) = q_k(x)$ almost surely. Overlap gives $p_{g,1}(x) \geq \varepsilon$ almost surely, and hence $\pi_{g,1} = E_P[p_{g,1}(X_i)] \geq \varepsilon > 0$. Consequently, iterated expectation gives

$$\begin{aligned} \frac{E_P[p_{g,1}(X_i)d_{k,c}(X_i)]}{\pi_{g,1}} &= \frac{E_P[\mathbf{1}\{G_i = g, S_i = 1\}q_k(X_i)]}{\pi_{g,1}} \\ &= E_Q[Y_{it}(g, 1) - Y_{it}(\infty, 1) \mid G_i = g, S_i = 1] \\ &= \theta_k(Q). \end{aligned}$$

This equality holds for every $c \in \mathcal{J}_k$, so averaging over c gives $\Theta_k(P) = \theta_k(Q)$.

It remains to compute the score mean. Conditional on (G_i, S_i, X_i) , each residual $\Delta Y_{i,k} - m_{a,s,k}(X_i)$ appearing in the score has mean zero. All residual summands therefore integrate to zero. Moreover,

$$E_P \left[\frac{\mathbf{1}\{G_i = g, S_i = 1\}}{\pi_{g,1}} \{d_{k,c}(X_i) - \Theta_k(P)\} \right] = \frac{E_P[p_{g,1}(X_i)d_{k,c}(X_i)]}{\pi_{g,1}} - \Theta_k(P) = 0.$$

The pairwise-score definition now implies $E_P[R_{k,c}(O_i; \eta_0)] = \Theta_k(P) = \theta_k(Q)$. Stacking these coordinate identities gives

$$\theta(Q) = \Theta(P), \quad E_P[R(O_i; \eta_0)] = B\Theta(P), \quad E_P[\Psi(O_i)] = 0.$$

Premultiplication by the prespecified vector ω gives

$$\tau_\omega(Q) = \omega' \theta(Q) = \omega' \Theta(P) = \Theta_\omega^{\text{agg}}(P).$$

Finally, when $g = 2$, the baseline is $g - 1 = 1$, so every change above is the observed change from period 1 to period t . $\square$

Justification. This published identification reduction fixes the exact score system whose cross-cell information is compared. *Gap.* Ortiz-Villavicencio and Sant'Anna prove the underlying pairwise DDD identification; the lemma is supporting structure and is not claimed as novel. *Consumer.* The joint GMM theorem uses the observable common-mean restriction $E_P[R] = B\Theta(P)$ while its Wald interval targets the causal aggregate $\tau_\omega(Q)$.

Lemma 2 (lem:shared-covariance-expansion). *For every $Q \in \mathcal{M}$ with observed-data law $P = Q_O$, $\Omega = \Omega^{\text{DDD}}$. In particular, the residual part of $\Omega_{(k,c),(\ell,d)}$ contains a primitive cell (a, s) if and only if that cohort-eligibility cell enters both pairwise scores; covariance across distinct observed cells vanishes by mutual exclusivity, while repeated outcome histories within a shared cell enter through $v_{a,s;k\ell}(X_i)$.*

Proof. Let $j = (k, c)$ and $q = (\ell, d)$, with $k = (g_k, t_k)$ and $\ell = (g_\ell, t_\ell)$. By the pairwise-score definition and Lemma 1, decompose the centered score as

$$\Psi_j = T_j + \sum_{a,s} U_j^{a,s},$$

where

$$T_j = \frac{\mathbf{1}\{G_i = g_k, S_i = 1\}}{\pi_{g_k,1}} \{d_{k,c}(X_i) - \Theta_k(P)\}$$

and

$$U_j^{a,s} = \mathbf{1}\{G_i = a, S_i = s\} L_{k,c}^{a,s}(X_i) \{\Delta Y_{i,k} - m_{a,s,k}(X_i)\}.$$

The sum ranges over the four cohort-eligibility cells entering coordinate j . Conditional residual centering gives

$$E_P[U_j^{a,s} \mid G_i, S_i, X_i] = 0.$$

It follows by iterated expectation that every covariance between a target-cell contrast term and a residual term is zero, including when the residual belongs to the target cell.

For the two target-cell terms, mutual exclusivity gives $T_j T_q = 0$ when $g_k \neq g_\ell$. When $g_k = g_\ell$, iterated expectation gives

$$E_P[T_j T_q] = E_P \left[\frac{p_{g_k,1}(X_i) \{d_{k,c}(X_i) - \Theta_k(P)\} \{d_{\ell,d}(X_i) - \Theta_\ell(P)\}}{\pi_{g_k,1} \pi_{g_\ell,1}} \right],$$

which is the first term in Definition 4.

For residual terms, $U_j^{a,s} U_q^{b,r} = 0$ identically unless $(a, s) = (b, r)$, again by mutual exclusivity of observed cohort-eligibility cells. When the same cell enters both scores, conditioning on $X_i = x$ gives

$$E_P[U_j^{a,s} U_q^{a,s} \mid X_i = x] = p_{a,s}(x) L_{k,c}^{a,s}(x) L_{\ell,d}^{a,s}(x) v_{a,s;k\ell}(x).$$

Summing over exactly the cells shared by the two coordinates and adding the target-cell term yields Ω_{jq}^{DDD} . Since this holds for every pair (j, q) ,

$$\Omega = E_P[\Psi(O_i) \Psi(O_i)'] = \Omega^{\text{DDD}}.$$

The same calculation proves the stated interpretation: distinct observed cells contribute no covariance, whereas two outcome changes observed in the same cell contribute their conditional covariance $v_{a,s;k\ell}(X_i)$. $\square$

Justification. This expansion is the DDD-specific step that converts generic systems covariance into shared eligible and ineligible comparison-cell information. *Gap.* Neither generic GMM nor seemingly unrelated regressions records the mutually exclusive cohort-eligibility cells and repeated panel outcome changes that determine staggered-DDD cross-block covariance. *Consumer.* The frontier diagnostic can be computed from cellwise residual covariances without treating comparison-cohort estimates as independent.

Lemma 3 (lem:cross-fitted-score-expansion). *Fix $F, T, \mathcal{K}$, and M . For every $Q \in \mathcal{M}$ with $P = Q_O$, suppose the off-fold learners satisfy foldwise independence, $\max_{f,a,s} \|\hat{p}_{a,s,-f} - p_{a,s}\|_{P_{X,4}} = O_P(a_n)$, $\max_{f,a,s,k} \|\hat{m}_{a,s,k,-f} - m_{a,s,k}\|_{P_{X,4}} = O_P(b_n)$, $a_n \rightarrow 0$, $b_n \rightarrow 0$, and $a_n b_n = o(n^{-1/2})$. These learner sequences and rates are theorem-level conditional hypotheses; no uniform learner-achievability claim over $\mathcal{M}$ is made. Four disjoint cells enter each retained pairwise score, so overlap and $0 < \varepsilon < 1/4$ give $p_{a,s}(x) \in [\varepsilon, 1 - 3\varepsilon] \subset [\varepsilon/2, 1 - \varepsilon/2]$ for every retained cell. Hence $C_\varepsilon(p_{a,s}) = p_{a,s}$, the clipping map is one-Lipschitz, and*

$$\|\tilde{p}_{a,s,-f} - p_{a,s}\|_{P_{X,4}} \leq \|\hat{p}_{a,s,-f} - p_{a,s}\|_{P_{X,4}} = O_P(a_n).$$

The primitive outcome fourth moment and overlap imply by conditional Jensen's inequality that every retained true regression is $L_4(P_X)$ -bounded and that $\|\Delta Y_{i,k} - m_{a,s,k}(X_i)\|_{P(\cdot|a,s),4} = O(1)$. The triangle inequality and the stated regression error rate then give $\max_{f,a,s,k} \|\hat{m}_{a,s,k,-f}\|_{P_{X,4}} = O_P(1)$; no separate true- or fitted-regression envelope is assumed. For each residual term, clipping, the Lipschitz bound for $(u,v) \mapsto u/v$ on $[\varepsilon/2, 1 - \varepsilon/2]^2$, and Hölder's inequality yield

$$\left\| \mathbf{1}_{a,s} \left(\frac{\tilde{p}_{g,1}}{\tilde{p}_{a,s}} - \frac{p_{g,1}}{p_{a,s}} \right) (\Delta Y_k - m_{a,s,k}) \right\|_{P,2} \leq C_\varepsilon (\|\tilde{p}_{g,1} - p_{g,1}\|_{P_{X,4}} + \|\tilde{p}_{a,s} - p_{a,s}\|_{P_{X,4}}) \|\Delta Y_k - m_{a,s,k}\|_{P(\cdot|a,s),4} = O_P(a_n),$$

and

$$\left\| \mathbf{1}_{a,s} \frac{\tilde{p}_{g,1}}{\tilde{p}_{a,s}} (\hat{m}_{a,s,k} - m_{a,s,k}) \right\|_{P,2} \leq C_\varepsilon \|\hat{m}_{a,s,k} - m_{a,s,k}\|_{P_{X,2}} = O_P(b_n).$$

The fitted target-contrast term is bounded by the sum of its four regression errors. Conditional score bias is bounded by $C_\varepsilon \max_{p,m} \|\hat{p} - p\|_{P_{X,2}} \|\hat{m} - m\|_{P_{X,2}} = O_P(a_n b_n) = o_P(n^{-1/2})$; foldwise independence makes every centered validation-fold plug-in fluctuation $o_P(1)$ after $n^{-1/2}$ summation. The empirical target share is root- n consistent, lies in the interior clipping region with probability tending to one, and its ratio expansion supplies exactly the target-share term in Ψ . Therefore, uniformly over the fixed score coordinates,

$$\sqrt{n}\{\hat{\mu} - B\Theta(P)\} = \frac{1}{\sqrt{n}} \sum_{i=1}^n \Psi(O_i) + o_P(1), \quad \max_{1 \leq j \leq M} \frac{1}{n} \sum_{i=1}^n \{\hat{\Psi}_{i,j} - \Psi_j(O_i)\}^2 = o_P(1).$$

Proof. Fix a coordinate $j = (k, c)$, write $k = (g, t)$, and abbreviate $I_{a,s} = \mathbf{1}\{G_i = a, S_i = s\}$. Constants below may depend on the fixed objects $F, T, \mathcal{K}, M, \varepsilon, C_4$, but not on n .

First derive the regression envelopes. The inequality

$$|\Delta Y_{i,k}|^4 \leq 8\{|Y_{it}|^4 + |Y_{i,g-1}|^4\} \leq 16 \max_{u \in \mathcal{T}} |Y_{iu}|^4$$

and the fourth-moment assumption imply $E_P |\Delta Y_{i,k}|^4 \leq 16C_4$. Conditional Jensen's inequality gives, on each retained cell,

$$|m_{a,s,k}(x)|^4 \leq E_P[|\Delta Y_{i,k}|^4 \mid G_i = a, S_i = s, X_i = x].$$

Because $p_{a,s}(x) \geq \varepsilon$, integration with respect to P_X yields

$$\begin{aligned} \|m_{a,s,k}\|_{P_{X,4}}^4 &\leq \int E_P[|\Delta Y_{i,k}|^4 \mid a, s, X_i = x] dP_X(x) \\ &\leq \varepsilon^{-1} E_P[I_{a,s} |\Delta Y_{i,k}|^4] \leq 16C_4/\varepsilon. \end{aligned}$$

The same calculation under the conditional law of a retained cell, together with $|u - v|^4 \leq 8(|u|^4 + |v|^4)$, gives

$$\|\Delta Y_{i,k} - m_{a,s,k}(X_i)\|_{P(\cdot|a,s),4} = O(1).$$

The regression error condition and the triangle inequality then imply, uniformly over the fixed folds and retained indices,

$$\|\widehat{m}_{a,s,k,-f}\|_{P_X,4} \leq \|m_{a,s,k}\|_{P_X,4} + \|\widehat{m}_{a,s,k,-f} - m_{a,s,k}\|_{P_X,4} = O_P(1).$$

Thus neither a true- nor a fitted-regression envelope is an extra hypothesis.

Next consider clipping. The four retained events $(g, 0), (g, 1), (c, 0), (c, 1)$ are distinct and belong to $\mathcal{C}_{\text{ret}}$. Hence $L_{\text{ret}} \geq 4$, and the overlap domain $0 < \varepsilon < L_{\text{ret}}^{-1}$ implies $0 < \varepsilon < 1/4$. Conditional overlap and disjointness further imply

$$\varepsilon \leq p_{a,s}(x) \leq 1 - 3\varepsilon \leq 1 - \varepsilon/2$$

for each of the four cells. Hence $C_\varepsilon(p_{a,s}) = p_{a,s}$. Projection onto an interval is one-Lipschitz, so fold uniformly

$$\|\widetilde{p}_{a,s,-f} - p_{a,s}\|_{P_X,4} \leq \|\widehat{p}_{a,s,-f} - p_{a,s}\|_{P_X,4} = O_P(a_n).$$

On $[\varepsilon/2, 1 - \varepsilon/2]^2$, the map $(u, v) \mapsto u/v$ is Lipschitz with a finite constant K_ε . Hence

$$\left\| \frac{\widetilde{p}_{g,1}}{\widetilde{p}_{a,s}} - \frac{p_{g,1}}{p_{a,s}} \right\|_{P_X,4} = O_P(a_n).$$

Hölder's inequality and the residual fourth-moment bound give

$$\begin{aligned} & \left\| I_{a,s} \left(\frac{\widetilde{p}_{g,1}}{\widetilde{p}_{a,s}} - \frac{p_{g,1}}{p_{a,s}} \right) \{ \Delta Y_{i,k} - m_{a,s,k}(X_i) \} \right\|_{P,2} \\ & \leq K_\varepsilon (\|\widetilde{p}_{g,1} - p_{g,1}\|_{P_X,4} + \|\widetilde{p}_{a,s} - p_{a,s}\|_{P_X,4}) \|\Delta Y_{i,k} - m_{a,s,k}(X_i)\|_{P(\cdot|a,s),4} = O_P(a_n), \end{aligned}$$

and bounded clipping gives

$$\left\| I_{a,s} \frac{\widetilde{p}_{g,1}}{\widetilde{p}_{a,s}} \{ \widehat{m}_{a,s,k} - m_{a,s,k} \} \right\|_{P,2} \leq K_\varepsilon \|\widehat{m}_{a,s,k} - m_{a,s,k}\|_{P_X,2} = O_P(b_n).$$

The fitted contrast error $\widehat{d}_{k,c} - d_{k,c}$ is a signed sum of four regression errors and is therefore $O_P(b_n)$ in $L_4(P_X)$.

Define the oracle numerator summand

$$A_j(O_i) = I_{g,1} d_{k,c}(X_i) + p_{g,1}(X_i) \sum_{a,s} \sigma_{a,s}^{g,c} \frac{I_{a,s}}{p_{a,s}(X_i)} \{ \Delta Y_{i,k} - m_{a,s,k}(X_i) \},$$

where the sum is over the four retained cells, and let $\widehat{A}_{i,j}$ be the fitted summand in Definition 7. The preceding bounds imply, conditionally on each training sample,

$$\|\widehat{A}_{i,j} - A_j(O_i)\|_{P,2} = O_P(a_n + b_n) = o_P(1).$$

There is also an exact orthogonality calculation. Put $e_{a,s} = \widehat{m}_{a,s,k} - m_{a,s,k}$. Conditional residual centering gives

$$E_P[\widehat{A}_{i,j} - A_j(O_i) \mid X_i = x] = \sum_{a,s} \sigma_{a,s}^{g,c} \left\{ p_{g,1}(x) - \widetilde{p}_{g,1}(x) \frac{p_{a,s}(x)}{\widetilde{p}_{a,s}(x)} \right\} e_{a,s}(x).$$

The bracket has $L_4(P_X)$ norm at most a constant times the relevant propensity errors. Hölder's inequality thus gives, fold uniformly,

$$\left| E_P[\widehat{A}_{i,j} - A_j(O_i) \mid \text{training sample}] \right| = O_P(a_n b_n) = o_P(n^{-1/2}).$$

Condition on the training sample for a validation fold. Foldwise independence and conditional Chebyshev's inequality show that the centered sum of $\hat{A}_{i,j} - A_j(O_i)$, divided by $\sqrt{n}$, has conditional variance bounded by a fixed multiple of $\|\hat{A}_j - A_j\|_{P,2}^2 = o_P(1)$. The preceding conditional-bias bound controls its mean. Since F is fixed, summing across folds gives

$$\frac{1}{\sqrt{n}} \sum_{i=1}^n \{\hat{A}_{i,j} - A_j(O_i)\} = o_P(1).$$

Let $\bar{I}_g = n^{-1} \sum_i I_{g,1}$ and $\pi = \pi_{g,1}$. Overlap gives $\pi \in [\varepsilon, 1 - 3\varepsilon]$. Bernoulli variance and Chebyshev's inequality give $\bar{I}_g - \pi = O_P(n^{-1/2})$; consequently target-share clipping is inactive with probability tending to one. By the explicit oracle formula in Definition 3, together with Lemma 1 and residual centering,

$$E_P A_j = \pi \Theta_k(P), \quad A_j(O_i) - \Theta_k(P) I_{g,1} = \pi \Psi_j(O_i).$$

On the event where clipping is inactive,

$$\begin{aligned} \sqrt{n}\{\hat{\mu}_j - \Theta_k(P)\} &= \frac{n^{-1/2} \sum_i \{A_j(O_i) - \Theta_k(P) I_{g,1}\}}{\bar{I}_g} + \frac{n^{-1/2} \sum_i \{\hat{A}_{i,j} - A_j(O_i)\}}{\bar{I}_g} \\ &= \frac{\pi}{\bar{I}_g} \frac{1}{\sqrt{n}} \sum_{i=1}^n \Psi_j(O_i) + o_P(1) \\ &= \frac{1}{\sqrt{n}} \sum_{i=1}^n \Psi_j(O_i) + o_P(1). \end{aligned}$$

The last equality uses $\bar{I}_g/\pi \rightarrow_P 1$ and $n^{-1/2} \sum_i \Psi_j(O_i) = O_P(1)$, the latter following from the finite second moment supplied by the derived residual bounds and overlap. Fixed M permits stacking:

$$\sqrt{n}\{\hat{\mu} - B\Theta(P)\} = \frac{1}{\sqrt{n}} \sum_{i=1}^n \Psi(O_i) + o_P(1).$$

Finally compare the fitted and oracle influence coordinates. On the same event, their difference is a finite sum of: the denominator error $\bar{I}_g^{-1} - \pi^{-1}$; the fitted-contrast error; the centering error $\hat{\mu}_j - \Theta_k(P)$; a propensity-ratio error times a true residual; and a bounded fitted ratio times a regression error. Conditional Markov's inequality, foldwise independence, and the bounds already proved yield

$$\frac{1}{n} \sum_{i=1}^n \{\hat{\Psi}_{i,j} - \Psi_j(O_i)\}^2 \leq O_P(1) \{(\bar{I}_g - \pi)^2 + |\hat{\mu}_j - \Theta_k(P)|^2 + a_n^2 + b_n^2\} = o_P(1).$$

Here the empirical second moments of oracle residual terms are $O_P(1)$, and the expansion just proved implies $\hat{\mu}_j - \Theta_k(P) = O_P(n^{-1/2})$. Taking the maximum over the fixed M coordinates proves the second assertion. $\square$

Justification. The lemma derives the specialized clipped ratio-adjusted staggered-DDD expansion and fitted-score L_2 convergence from primitive moments, overlap, and supplied learner rates. *Gap.* Generic cross-fitting results do not by themselves provide this score-specific calculation or uniform learner availability; the explicit pairwise-score definition is now declared in the proof spine. *Consumer.* The covariance-consistency lemma and joint limit theorem use the resulting asymptotic linearity and empirical fitted-score control.

Lemma 4 (lem:fitted-covariance-consistency). *Fix F , T , $\mathcal{K}$, and M . For every $Q \in \mathcal{M}$ with $P = Q_O$, impose the theorem-level foldwise independence, propensity and regression $L_4(P_X)$ rates, and product-rate condition. No separate true- or fitted-regression envelope is imposed: the score-expansion lemma derives the required envelopes from the primitive outcome fourth moment, overlap, and the regression error rate. That lemma gives $D_{i,j} = \hat{\Psi}_{i,j} - \Psi_j(O_i)$ with $\max_j n^{-1} \sum_i D_{i,j}^2 = o_P(1)$. For every $1 \leq j, q \leq M$,*

$$\hat{\Omega}_{jq}^{\text{raw}} - \Omega_{jq} = \left\{ n^{-1} \sum_{i=1}^n \Psi_j(O_i) \Psi_q(O_i) - \Omega_{jq} \right\} + n^{-1} \sum_{i=1}^n D_{i,j} \hat{\Psi}_{i,q} + n^{-1} \sum_{i=1}^n \Psi_j(O_i) D_{i,q}.$$

The oracle empirical-covariance fluctuation is $o_P(1)$ by i.i.d. sampling and the derived fourth-moment bound. Cauchy-Schwarz gives

$$\begin{aligned} |\widehat{\Omega}_{jq}^{\text{raw}} - \Omega_{jq}| &\leq \left| n^{-1} \sum_{i=1}^n \Psi_j(O_i) \Psi_q(O_i) - \Omega_{jq} \right| \\ &\quad + \left(n^{-1} \sum_i D_{i,j}^2 \right)^{1/2} \left(n^{-1} \sum_i \widehat{\Psi}_{i,q}^2 \right)^{1/2} + \left(n^{-1} \sum_i \Psi_j(O_i)^2 \right)^{1/2} \left(n^{-1} \sum_i D_{i,q}^2 \right)^{1/2} = o_P(1). \end{aligned}$$

Since M is fixed, $\|A\|_{\text{op}} \leq M \max_{j,q} |A_{jq}|$ passes entrywise convergence to operator norm. Because $\Omega \in \mathbb{S}_+^M$ and positive-semidefinite projection is nonexpansive in Frobenius norm,

$$\|\widehat{\Omega} - \Omega\|_{\text{op}} = o_P(1), \quad \|\widehat{\Omega}_\lambda - \Omega\|_{\text{op}} \leq \|\widehat{\Omega} - \Omega\|_{\text{op}} + \lambda_n = o_P(1).$$

Proof. Set $D_{i,j} = \widehat{\Psi}_{i,j} - \Psi_j(O_i)$. Lemma 3, which derives rather than assumes the regression envelopes, gives

$$\max_{1 \leq j \leq M} \frac{1}{n} \sum_{i=1}^n D_{i,j}^2 = o_P(1).$$

The same derived moment bounds and overlap imply $E_P |\Psi_j(O_i)|^4 < \infty$ for every fixed coordinate: the target term is an indicator times an $L_4(P_X)$ contrast with bounded denominator, and every residual term has a bounded inverse-propensity loading and a finite cell-conditional fourth moment. Hence $E_P [\Psi_j(O_i)^2 \Psi_q(O_i)^2] < \infty$. By i.i.d. sampling and Chebyshev's inequality,

$$\frac{1}{n} \sum_{i=1}^n \Psi_j(O_i) \Psi_q(O_i) - \Omega_{jq} = o_P(1).$$

Using $\widehat{\Psi}_j = \Psi_j + D_j$, the identity

$$\widehat{\Omega}_{jq}^{\text{raw}} - \Omega_{jq} = \left\{ \frac{1}{n} \sum_i \Psi_j(O_i) \Psi_q(O_i) - \Omega_{jq} \right\} + \frac{1}{n} \sum_i D_{i,j} \widehat{\Psi}_{i,q} + \frac{1}{n} \sum_i \Psi_j(O_i) D_{i,q}$$

is exact. Cauchy-Schwarz yields

$$\begin{aligned} |\widehat{\Omega}_{jq}^{\text{raw}} - \Omega_{jq}| &\leq \left| \frac{1}{n} \sum_i \Psi_j(O_i) \Psi_q(O_i) - \Omega_{jq} \right| \\ &\quad + \left(\frac{1}{n} \sum_i D_{i,j}^2 \right)^{1/2} \left(\frac{1}{n} \sum_i \widehat{\Psi}_{i,q}^2 \right)^{1/2} \\ &\quad + \left(\frac{1}{n} \sum_i \Psi_j(O_i)^2 \right)^{1/2} \left(\frac{1}{n} \sum_i D_{i,q}^2 \right)^{1/2}. \end{aligned}$$

Again by i.i.d. sampling, $n^{-1} \sum_i \Psi_j(O_i)^2 = O_P(1)$, while

$$\frac{1}{n} \sum_i \widehat{\Psi}_{i,q}^2 \leq 2 \frac{1}{n} \sum_i \Psi_q(O_i)^2 + 2 \frac{1}{n} \sum_i D_{i,q}^2 = O_P(1).$$

Every term in the preceding bound is therefore $o_P(1)$. Since M is fixed,

$$\|\widehat{\Omega}^{\text{raw}} - \Omega\|_{\text{op}} \leq M \max_{j,q} |\widehat{\Omega}_{jq}^{\text{raw}} - \Omega_{jq}| = o_P(1).$$

Let Π denote Frobenius projection onto the closed convex cone $\mathbb{S}_+^M$. The projection variational inequalities imply

$$\|\Pi(A) - \Pi(C)\|_F^2 \leq \langle \Pi(A) - \Pi(C), A - C \rangle_F \leq \|\Pi(A) - \Pi(C)\|_F \|A - C\|_F.$$

Take $C = \Omega$. Since $\Omega \in \mathbb{S}_+^M$, $\Pi(C) = C$, and fixed dimensionality gives

$$\|\widehat{\Omega} - \Omega\|_{\text{op}} \leq \|\widehat{\Omega} - \Omega\|_F \leq \|\widehat{\Omega}^{\text{raw}} - \Omega\|_F = o_P(1).$$

Finally, $\lambda_n = n^{-1/4} \rightarrow 0$, and therefore

$$\|\widehat{\Omega}_\lambda - \Omega\|_{\text{op}} \leq \|\widehat{\Omega} - \Omega\|_{\text{op}} + \lambda_n = o_P(1).$$

□

Justification. Derives consistency of the exact projected and ridged covariance estimator after the redundant envelope assumptions are removed. *Gap.* The supporting probability argument is specialized to the displayed cross-fitted score vector with shared units. *Consumer.* Permits continuous inversion for the joint and componentwise feasible loadings.

Theorem 1 (thm:joint-system-limit). *Fix $F, T, \emptyset \neq \mathcal{K} \subseteq \{(g, t) : 2 \leq g \leq t \leq T\}$, and M . Fix a law $Q \in \mathcal{M}$, let $P = Q_O$, and fix supplied sequences of off-fold learners satisfying foldwise independence, $\max_{f,a,s} \|\widehat{p}_{a,s,-f} - p_{a,s}\|_{P_{X,4}} = O_P(a_n)$, $\max_{f,a,s,k} \|\widehat{m}_{a,s,k,-f} - m_{a,s,k}\|_{P_{X,4}} = O_P(b_n)$, $a_n \rightarrow 0$, $b_n \rightarrow 0$, and $a_n b_n = o(n^{-1/2})$. All stochastic orders, limits, and probabilities below are pointwise for this fixed pair consisting of Q and the supplied learner sequences. No existence of such learners uniformly over $\mathcal{M}$, and no uniform coverage over $\mathcal{M}$, is asserted. No fitted-regression envelope is separately assumed, because it follows from the primitive outcome fourth moment, overlap, and the regression error rate. Use the deterministic ridge $\lambda_n = n^{-1/4} > 0$. The everywhere-defined estimator in Definition 7 then satisfies*

$$\sqrt{n}\{\widehat{\tau}_{\text{joint}} - \Theta_\omega^{\text{agg}}(P)\} = \frac{1}{\sqrt{n}} \sum_{i=1}^n (b^{\text{joint}})' \Psi(O_i) + o_P(1) \rightsquigarrow N(0, V_{\text{joint}}), \quad \|\widehat{\Omega} - \Omega\|_{\text{op}} = o_P(1).$$

Moreover, b^{joint} is the unique minimizer of $b' \Omega b$ over constant $b \in \mathbb{R}^M$ satisfying $B'b = \omega$, $\widehat{b}^{\text{joint}} \rightarrow_P b^{\text{joint}}$, $\widehat{V}_{\text{joint}} \rightarrow_P V_{\text{joint}} > 0$, and

$$\lim_{n \rightarrow \infty} P\{\tau_\omega(Q) \in \mathcal{C}_{n,1-\alpha}\} = 1 - \alpha.$$

This is conditional pointwise inference and finite-dimensional constrained-GMM optimality within the constant span of the retained pairwise staggered-DDD scores, supported by the model-specific clipped cross-fitted score expansion. It is not an efficient influence function or a semiparametric efficiency bound for the full conditional-moment model. If a target coordinate has $g = 2$, its score and fitted score use $\Delta Y_{i,k} = Y_{it} - Y_{i1}$. If $\mathcal{K} = \{k\}$, the joint loading, estimator, and variance reduce to the published within- k comparison-cohort GMM objects. The zero-variance convention in Definition 8 totalizes the finite-sample interval and is asymptotically inactive.

Proof. Fix the pair consisting of Q and the supplied learner sequences. Because a retained score contains the four distinct cells $(g, 0), (g, 1), (c, 0), (c, 1)$, the retained-cell definition gives $L_{\text{ret}} \geq 4$. The overlap domain therefore implies $0 < \varepsilon < 1/4$, and overlap with the other three cells gives the per-score upper bound $p_{a,s} \leq 1 - 3\varepsilon$. Hence the clipping and moment arguments in Lemmas 3 and 4 apply pointwise to this fixed pair and give

$$Z_n := \sqrt{n}\{\widehat{\mu} - B\Theta(P)\} = \frac{1}{\sqrt{n}} \sum_{i=1}^n \Psi(O_i) + o_P(1), \quad \widehat{\Omega} \rightarrow_P \Omega, \quad \widehat{\Omega}_\lambda \rightarrow_P \Omega.$$

These conclusions derive the needed fitted-score and covariance behavior from the displayed learner-rate hypotheses; they do not construct the learners.

Each column of B has a nonempty row support, and the supports of distinct columns are disjoint. Thus B has full column rank. The score-spectrum assumption makes Ω positive definite, so $B'\Omega^{-1}B$ is positive definite. Continuity of matrix inversion at positive-definite matrices yields

$$\widehat{b}^{\text{joint}} = \widehat{\Omega}_\lambda^{-1} B (B' \widehat{\Omega}_\lambda^{-1} B)^{-1} \omega \rightarrow_P \Omega^{-1} B (B' \Omega^{-1} B)^{-1} \omega = b^{\text{joint}}.$$

For every sample, $\widehat{\Omega}_\lambda$ is positive definite because $\widehat{\Omega}$ is positive semidefinite and $\lambda_n > 0$. The same full-rank argument makes $B' \widehat{\Omega}_\lambda^{-1} B$ positive definite. Together with propensity and target-share clipping, this proves that every denominator and inverse in $\widehat{\tau}_{\text{joint}}$ is defined for every realized sample. Direct multiplication also gives

$$B' \widehat{b}^{\text{joint}} = \omega.$$

Therefore

$$\begin{aligned} \sqrt{n} \{ \widehat{\tau}_{\text{joint}} - \Theta_\omega^{\text{agg}}(P) \} &= (\widehat{b}^{\text{joint}})' \sqrt{n} \{ \widehat{\mu} - B\Theta(P) \} \\ &= (b^{\text{joint}})' \frac{1}{\sqrt{n}} \sum_{i=1}^n \Psi(O_i) + o_P(1), \end{aligned}$$

because $Z_n = O_P(1)$ and $\widehat{b}^{\text{joint}} - b^{\text{joint}} = o_P(1)$.

Put $Z_i = (b^{\text{joint}})' \Psi(O_i)$. By independent sampling and Lemma 1, the Z_i are independent and identically distributed with mean zero and variance

$$E_P[Z_i^2] = (b^{\text{joint}})' \Omega b^{\text{joint}} = V_{\text{joint}}.$$

The fourth-moment and overlap derivations in Lemma 3 imply $E_P|\Psi_j(O_i)|^4 < \infty$ for every fixed coordinate, and fixed M therefore gives $E_P|Z_i|^4 < \infty$. For $Z_{n,i} = Z_i/\sqrt{n}$,

$$\sum_{i=1}^n \text{Var}(Z_{n,i}) = V_{\text{joint}},$$

and, for every $r > 0$,

$$\sum_{i=1}^n E_P[Z_{n,i}^2 \mathbf{1}\{|Z_{n,i}| > r\}] = E_P[Z_i^2 \mathbf{1}\{|Z_i| > r\sqrt{n}\}] \leq \frac{E_P[Z_i^4]}{r^2 n} \rightarrow 0.$$

The Lindeberg–Feller theorem in Lemma 5 gives

$$\frac{1}{\sqrt{n}} \sum_{i=1}^n Z_i \rightsquigarrow N(0, V_{\text{joint}}).$$

To absorb the remainder directly, if $r_n = o_P(1)$, then for every $h > 0$ and every continuity point x of the limiting normal distribution,

$$P(X_n \leq x - h) - P(|r_n| > h) \leq P(X_n + r_n \leq x) \leq P(X_n \leq x + h) + P(|r_n| > h),$$

where $X_n = n^{-1/2} \sum_i Z_i$. Taking limits and then $h \downarrow 0$ proves the asserted limit law.

It remains to establish the finite-span optimum. Consider the Lagrangian

$$\mathcal{L}(b, \lambda) = b' \Omega b - 2\lambda'(B'b - \omega).$$

Its first-order conditions are $b = \Omega^{-1}B\lambda$ and $B'b = \omega$, whence

$$\lambda = (B'\Omega^{-1}B)^{-1}\omega, \quad b = \Omega^{-1}B(B'\Omega^{-1}B)^{-1}\omega = b^{\text{joint}}.$$

Strict convexity of $b \mapsto b'\Omega b$ makes this feasible solution the unique minimizer. Because $\mathbf{1}'\omega = 1$, the constraint $B'b^{\text{joint}} = \omega$ implies $b^{\text{joint}} \neq 0$. The spectrum bound then gives

$$V_{\text{joint}} = (b^{\text{joint}})'\Omega b^{\text{joint}} \geq \kappa \|b^{\text{joint}}\|_2^2 > 0.$$

The already-proved convergences of the loading and covariance imply

$$\widehat{V}_{\text{joint}} = (\widehat{b}^{\text{joint}})'\widehat{\Omega}\widehat{b}^{\text{joint}} \rightarrow_P V_{\text{joint}}.$$

Lemma 1 gives $\tau_\omega(Q) = \Theta_\omega^{\text{agg}}(P)$. Hence

$$\frac{\sqrt{n}\{\widehat{\tau}_{\text{joint}} - \tau_\omega(Q)\}}{\sqrt{\widehat{V}_{\text{joint}}}} \rightsquigarrow N(0, 1).$$

Indeed, $\sqrt{V_{\text{joint}}/\widehat{V}_{\text{joint}}} - 1 = o_P(1)$, and multiplying this error by the tight unstudentized standardized statistic is $o_P(1)$; the same distribution-function sandwich used above transfers the limit. Continuity of the standard normal distribution at $\pm z_{1-\alpha/2}$ now proves

$$\lim_{n \rightarrow \infty} P\{\tau_\omega(Q) \in \mathcal{C}_{n,1-\alpha}\} = 1 - \alpha.$$

Since $V_{\text{joint}} > 0$ and $\widehat{V}_{\text{joint}} \rightarrow_P V_{\text{joint}}$, the event $\widehat{V}_{\text{joint}} = 0$ has probability tending to zero, so the finite-sample zero-variance convention is asymptotically inactive.

When $g = 2$, the outcome-change definition gives $\Delta Y_{i,k} = Y_{it} - Y_{i1}$ in both oracle and fitted scores. If $\mathcal{K} = \{k\}$, then $B = \mathbf{1}$ and $\omega_k = 1$, so

$$b^{\text{joint}} = \frac{\Omega^{-1}\mathbf{1}}{\mathbf{1}'\Omega^{-1}\mathbf{1}} = b^{\text{sep}}.$$

The feasible loading, estimator, and variance formulas reduce accordingly. Every conclusion is pointwise for the fixed law/learner pair and confined to the finite constant retained-score span; no tangent-space or full semiparametric-efficiency assertion has been used. $\square$

Justification. The theorem gives an everywhere-defined feasible joint constant-span estimator, pointwise limit law, variance consistency, and Wald coverage for the identified causal aggregate. *Gap.* Relative to separate within-block GMM and fixed-weight stacked-DDD inference, it estimates the joint covariance-optimal loading over the complete retained score vector; it does not claim a full semiparametric bound or novelty for shared-unit covariance accounting. *Consumer.* Applied researchers with supplied nuisance learners meeting the displayed rates obtain a fully specified point estimator and pointwise Wald interval.

Theorem 2 (thm:separate-joint-frontier). *Fix F , T , $\emptyset \neq \mathcal{K} \subseteq \{(g, t) : 2 \leq g \leq t \leq T\}$, and M . The aggregate formed from the Ortiz-Villavicencio–Sant’Anna componentwise-GMM ATT functionals using the prespecified weights is joint-optimal in the constant retained-score span if and only if $N'\Omega b^{\text{sep}} = 0$, equivalently if and only if each block of Ωb^{sep} is constant across the comparison cohorts in that block. For every $Q \in \mathcal{M}$, with $P = Q_O$, the population identity*

$$V_{\text{sep}} - V_{\text{joint}} = (N'\Omega b^{\text{sep}})'(N'\Omega N)^{-1}(N'\Omega b^{\text{sep}}) = (b^{\text{sep}} - b^{\text{joint}})'\Omega(b^{\text{sep}} - b^{\text{joint}})$$

holds without learner conditions, with the zero-dimensional quadratic interpreted as zero. Suppose in addition that the i.i.d. sampling, fixed-fold sample splitting, propensity and regression $L_4(P_X)$ rates, product-rate condition, and deterministic ridge condition of Theorem 1 all hold. The learner sequences and rates are theorem-level conditional hypotheses, and no uniform learner-achievability claim over $\mathcal{M}$ is made; no separate fitted-regression envelope is required. Then $\hat{\tau}_{\text{joint}}$ and $\hat{\tau}_{\text{sep}}$ have asymptotic variances V_{joint} and V_{sep} , respectively, so the joint estimator has strictly smaller asymptotic variance exactly when the net cross-block covariance contribution is nonconstant across comparison cohorts within at least one group-event block. Moreover, $\hat{\Delta} \geq 0$ for every realized sample and $\hat{\Delta} \rightarrow_P V_{\text{sep}} - V_{\text{joint}}$. Thus $\hat{\Delta}$ is a consistent descriptive diagnostic of the constant retained-score-span efficiency loss; no convergence rate, test, confidence interval, or full semiparametric-efficiency claim is made. For $g = 2$, all covariance entries use the defined baseline period 1. If $\mathcal{K} = \{k\}$, then $N'\Omega b^{\text{sep}} = 0$, $b^{\text{joint}} = b^{\text{sep}}$, and $V_{\text{sep}} - V_{\text{joint}} = 0$, reproducing within-cell GMM optimality.

Proof. Fix $Q \in \mathcal{M}$. Positive definiteness of Ω implies that every principal block Ω_{kk} is positive definite. Hence

$$\mathbf{1}'(b^{\text{sep}})_k = \omega_k, \quad \Omega_{kk}(b^{\text{sep}})_k = \frac{\omega_k}{\mathbf{1}'\Omega_{kk}^{-1}\mathbf{1}}\mathbf{1}.$$

Consequently $B'b^{\text{sep}} = \omega$, and the within-block contribution is constant across the comparison coordinates in block k .

The columns of $N_k = [e_1 - e_{J_k}, \dots, e_{J_k-1} - e_{J_k}]$, where $J_k = |\mathcal{J}_k|$, form a basis of the zero-sum subspace of $\mathbb{R}^{J_k}$. Thus N has full column rank and

$$\text{col}(N) = \ker(B').$$

Every feasible loading has a unique representation $b = b^{\text{sep}} + N\gamma$. If N has positive column dimension, put

$$H = N'\Omega N, \quad r = N'\Omega b^{\text{sep}}.$$

Then H is positive definite, and completing the square gives

$$\begin{aligned} (b^{\text{sep}} + N\gamma)'\Omega(b^{\text{sep}} + N\gamma) &= V_{\text{sep}} + 2\gamma'r + \gamma'H\gamma \\ &= V_{\text{sep}} - r'H^{-1}r + (\gamma + H^{-1}r)'H(\gamma + H^{-1}r). \end{aligned}$$

The unique constrained minimizer supplied by this completed square is

$$b^* = b^{\text{sep}} - NH^{-1}r.$$

To identify this minimizer without any learner conditions, let

$$C = B'\Omega^{-1}B.$$

The disjoint nonempty row supports of the columns of B make B full column rank, and positive definiteness of Ω therefore makes C positive definite. Definition 6 gives

$$b^{\text{joint}} = \Omega^{-1}BC^{-1}\omega, \quad B'b^{\text{joint}} = CC^{-1}\omega = \omega.$$

For every $h \in \ker(B')$,

$$h'\Omega b^{\text{joint}} = h'BC^{-1}\omega = (B'h)'C^{-1}\omega = 0.$$

Thus, for every feasible $b = b^{\text{joint}} + h$,

$$b'\Omega b = (b^{\text{joint}})'\Omega b^{\text{joint}} + h'\Omega h,$$

where the cross term vanishes. Positive definiteness of Ω makes the last term strictly positive unless $h = 0$, so b^{joint} is the unique minimizer over $\{b : B'b = \omega\}$. It must therefore equal the already-derived completed-square minimizer b^* . Consequently

$$b^{\text{sep}} - b^{\text{joint}} = N(N'\Omega N)^{-1}N'\Omega b^{\text{sep}},$$

and

$$\begin{aligned} V_{\text{sep}} - V_{\text{joint}} &= (N' \Omega b^{\text{sep}})' (N' \Omega N)^{-1} (N' \Omega b^{\text{sep}}) \\ &= (b^{\text{sep}} - b^{\text{joint}})' \Omega (b^{\text{sep}} - b^{\text{joint}}). \end{aligned}$$

If N has no columns, the feasible set is the singleton $\{b^{\text{sep}}\}$, and the same assertions hold with the zero-dimensional quadratic interpreted as zero.

Positive definiteness of H shows that the gap vanishes exactly when $N' \Omega b^{\text{sep}} = 0$. For any conformably partitioned vector v , the coordinates of $N'_k v_k$ are $v_{k,r} - v_{k,J_k}$, $r < J_k$. Hence $N'v = 0$ exactly when every block of v is constant across its comparison cohorts. Taking $v = \Omega b^{\text{sep}}$ gives the stated criterion. Moreover,

$$(\Omega b^{\text{sep}})_k = \Omega_{kk} (b^{\text{sep}})_k + \sum_{\ell \neq k} \Omega_{k\ell} (b^{\text{sep}})_\ell.$$

The first term is constant within block k . Thus strict gain occurs exactly when the net cross-block covariance contribution is nonconstant across comparison cohorts in some block.

Now impose the sampling, learner-rate, product-rate, and ridge conditions. Lemmas 3 and 4 give

$$Z_n := \sqrt{n} \{\hat{\mu} - B\Theta(P)\} = \frac{1}{\sqrt{n}} \sum_{i=1}^n \Psi(O_i) + o_P(1), \quad \hat{\Omega}_\lambda \rightarrow_P \Omega, \quad \hat{\Omega} \rightarrow_P \Omega.$$

Since each Ω_{kk} is positive definite, continuity of block inversion and the loading definitions imply $\hat{b}^{\text{sep}} \rightarrow_P b^{\text{sep}}$. Theorem 1 gives $\hat{b}^{\text{joint}} \rightarrow_P b^{\text{joint}}$ and the asserted joint limit law.

The fitted separate loading satisfies $B' \hat{b}^{\text{sep}} = \omega$ exactly. Hence

$$\begin{aligned} \sqrt{n} \{\hat{\tau}_{\text{sep}} - \Theta_\omega^{\text{agg}}(P)\} &= (\hat{b}^{\text{sep}})' Z_n \\ &= \frac{1}{\sqrt{n}} \sum_{i=1}^n (b^{\text{sep}})' \Psi(O_i) + o_P(1). \end{aligned}$$

Let $Z_i^{\text{sep}} = (b^{\text{sep}})' \Psi(O_i)$. Independent sampling and Lemma 1 give independent identically distributed centered summands with variance V_{sep} . The fourth-moment and overlap bounds used in Lemma 3 imply $E_P |Z_i^{\text{sep}}|^4 < \infty$. Thus, for every $r > 0$,

$$\begin{aligned} \sum_{i=1}^n E_P \left[\frac{(Z_i^{\text{sep}})^2}{n} \mathbf{1}_{\{|Z_i^{\text{sep}}| > r\sqrt{n}\}} \right] \\ = E_P [(Z_i^{\text{sep}})^2 \mathbf{1}_{\{|Z_i^{\text{sep}}| > r\sqrt{n}\}}] \leq \frac{E_P [(Z_i^{\text{sep}})^4]}{r^2 n} \rightarrow 0. \end{aligned}$$

Lemma 5 gives

$$\sqrt{n} \{\hat{\tau}_{\text{sep}} - \Theta_\omega^{\text{agg}}(P)\} \rightsquigarrow N(0, V_{\text{sep}}).$$

Together with Theorem 1 and Lemma 1, this proves that the two causal estimators have asymptotic variances V_{sep} and V_{joint} .

For every sample, positive semidefiniteness of $\hat{\Omega}$ gives $\hat{\Delta} \geq 0$. The loading and covariance convergences and continuity of the quadratic form give

$$\hat{\Delta} \rightarrow_P (b^{\text{sep}} - b^{\text{joint}})' \Omega (b^{\text{sep}} - b^{\text{joint}}) = V_{\text{sep}} - V_{\text{joint}}.$$

This proves descriptive consistency, not a rate or test for the gap.

Finally, if $\mathcal{K} = \{k\}$, then $\omega_k = 1$, $B = \mathbf{1}$, and $\Omega = \Omega_{kk}$. Thus

$$b^{\text{sep}} = \frac{\Omega^{-1} \mathbf{1}}{\mathbf{1}' \Omega^{-1} \mathbf{1}} = b^{\text{joint}}, \quad N' \Omega b^{\text{sep}} = 0.$$

The gap is zero. For $g = 2$, the outcome-change definition gives $\Delta Y_{i,k} = Y_{it} - Y_{i1}$, so every relevant covariance uses baseline period 1. $\square$

Justification. The theorem exposes the exact if-and-only-if frontier, quadratic variance gap, and consistent descriptive diagnostic after the DDD-specific shared-cell covariance map supplies the full covariance matrix. *Gap.* Ortiz-Villavicencio–Sant’Anna provide the separate blockwise benchmark and Hsieh provides fixed-weight shared-unit inference; neither provides the joint optimization or equality frontier. Caron (2025) derives a semiparametric efficiency bound and efficient estimators for a different parameter, the causal difference in subgroup ATTs under heterogeneous treatment effects. That result does not subsume this frontier for a prespecified staggered-panel aggregate of eligible-cohort effects, which is deliberately restricted to constant combinations of the retained pairwise-score vector. Generic constrained-GMM/SUR minimization is prior algebra. *Consumer.* Researchers can determine exactly when separate aggregation loses constant-score-span precision and report $\hat{\Delta}$ descriptively without a test or full-efficiency interpretation; this comparison is not a semiparametric efficiency claim.

Lemma 5 (lem:lindeberg-feller-central-limit). *For every triangular array of independent finite-variance real random variables $Z_{n,1}, \dots, Z_{n,n}$, if $\sum_{i=1}^n \text{Var}(Z_{n,i}) \rightarrow V$ and, for every $r > 0$, $\sum_{i=1}^n E_P[Z_{n,i}^2 \mathbf{1}\{|Z_{n,i}| > r\}] \rightarrow 0$, then $\sum_{i=1}^n \{Z_{n,i} - E_P Z_{n,i}\} \rightsquigarrow N(0, V)$. This is the scalar $k_n = n$ specialization of van der Vaart (1998, Proposition 2.27).*

Justification. Supplies the classical scalar triangular-array central limit theorem used in the limit-law arguments. *Gap.* This is cited substrate, not a novel result; the visible statement identifies van der Vaart (1998, Proposition 2.27). *Consumer.* The limit-law proofs verify its variance and Lindeberg conditions for scalar influence sums.

Theorem 3 (thm:labelled-incidence-classification-and-strict-gap). *Because $\mathfrak{H}(P)$ retains $\mathfrak{C}(P) = \Omega(P)$, N , ω , the comparison labels, and the covariance data, the map $[\mathfrak{H}(P)] \mapsto N' \mathfrak{C}(P) b^{\text{sep}}(P)$ is evaluable and constant on label-preserving equivalence classes. Consequently the classes admit the bookkeeping partition*

$$\mathcal{E}_0 = \{[\mathfrak{H}(P)] : N' \mathfrak{C}(P) b^{\text{sep}}(P) = 0\}, \quad \mathcal{E}_+ = \{[\mathfrak{H}(P)] : N' \mathfrak{C}(P) b^{\text{sep}}(P) \neq 0\}.$$

For invariants induced by compatible laws in $\mathcal{M}$, the joint variance gain is zero on $\mathcal{E}_0$ and strictly positive on $\mathcal{E}_+$. This partition is an evaluable restatement of the separate-joint frontier, not an independent or explanatory classification.

The strict class is nonempty even in the smallest architecture. Take $T = 3$, $\mathcal{K} = \{(2, 2), (2, 3)\}$, $\mathcal{J}_{(2,2)} = \{3, \infty\}$, $\mathcal{J}_{(2,3)} = \{\infty\}$, $\varepsilon = 1/8$, $C_4 = 1$, and $\kappa = 1/8$. For any prespecified $\omega = (\omega_1, \omega_2)'$ with $\omega_1 + \omega_2 = 1$ and $\omega_2 \neq 0$, there is a bounded compatible full-data law $Q \in \mathcal{M}$ for which, in the coordinate order $((2, 2), 3), ((2, 2), \infty), ((2, 3), \infty)$,

$$\Omega = \begin{pmatrix} 3/8 & 3/16 & 0 \\ 3/16 & 3/8 & 3/32 \\ 0 & 3/32 & 3/8 \end{pmatrix}, \quad b^{\text{sep}} = (\omega_1/2, \omega_1/2, \omega_2)',$$

and

$$V_{\text{sep}} - V_{\text{joint}} = \frac{3\omega_2^2}{128} > 0.$$

Indeed this witness is one point of the open nonzero-effect family in Theorem 4.

Proof. Let $\mathfrak{H}(P)$ be a labelled invariant in the stated domain. Definition 9 retains the labelled coordinate blocks, $\mathfrak{C}(P) = \Omega(P)$, N , and ω . Positive definiteness of Ω , which is part of the law class through the score-spectrum condition, implies positive definiteness of every principal block Ω_{kk} . Therefore

$$(b^{\text{sep}})_k = \omega_k \frac{\Omega_{kk}^{-1} \mathbf{1}}{\mathbf{1}' \Omega_{kk}^{-1} \mathbf{1}}$$

is well defined and evaluable: the inverse exists and the denominator is strictly positive. Stacking these finitely many blocks and carrying out finite matrix multiplication evaluates

$$\Phi(\mathfrak{H}(P)) := N' \mathfrak{C}(P) b^{\text{sep}}(P).$$

We next verify invariance. After identifying the labels under a label-preserving equivalence, let Π be the induced block-preserving coordinate permutation. The defining equivalence gives

$$\tilde{\mathfrak{C}} = \Pi \mathfrak{C} \Pi', \quad \tilde{N} = \Pi N, \quad \tilde{\omega} = \omega.$$

Blockwise, permutation conjugation commutes with inversion and $\Pi \mathbf{1} = \mathbf{1}$. Hence the componentwise loading formula gives

$$\tilde{b}^{\text{sep}} = \Pi b^{\text{sep}}.$$

Consequently

$$\tilde{N}' \tilde{\mathfrak{C}} \tilde{b}^{\text{sep}} = N' \Pi' \Pi \mathfrak{C} \Pi' \Pi b^{\text{sep}} = N' \mathfrak{C} b^{\text{sep}}.$$

Thus Φ is constant on label-preserving equivalence classes. Every finite vector is either zero or nonzero, so $\mathcal{E}_0$ and $\mathcal{E}_+$ are well defined, disjoint, and exhaustive.

Now restrict attention to an invariant induced by a compatible law $Q \in \mathcal{M}$, and write

$$u = N' \Omega b^{\text{sep}}, \quad H = N' \Omega N.$$

Theorem 2 gives

$$V_{\text{sep}} - V_{\text{joint}} = u' H^{-1} u$$

when N has columns, with the zero-dimensional quadratic interpreted as zero otherwise. If N has columns, its full column rank and positive definiteness of Ω imply, for every nonzero vector z ,

$$z' H z = (Nz)' \Omega (Nz) > 0.$$

Thus H is positive definite. It follows that the variance gain is zero exactly when $u = 0$, and it is strictly positive exactly when $u \neq 0$. This establishes the asserted bookkeeping partition and also shows that it is precisely a restatement of the separate-joint frontier.

It remains to prove that the strict class is nonempty with the displayed arithmetic. Lemma 6 shows that the retained cohort-eligibility cells in the proposed architecture are the six pairs obtained from cohort labels 2, 3, and ∞ and eligibility values 0 and 1. Hence $L_{\text{ret}} = 6$, and

$$0 < \frac{1}{8} < \frac{1}{6} = L_{\text{ret}}^{-1}.$$

Set

$$\rho = \frac{1}{2}, \quad \delta = \frac{1}{16}.$$

These choices obey

$$0 < |\rho| < \frac{\sqrt{7}}{3}, \quad 0 < |\delta| < \frac{1}{8}.$$

Theorem 4 therefore supplies a bounded compatible law $Q_{1/2,1/16} \in \mathcal{M}$, with nonzero constant post-adoption effects, for which

$$\Omega = \begin{pmatrix} 3/8 & 3/16 & 0 \\ 3/16 & 3/8 & 3/32 \\ 0 & 3/32 & 3/8 \end{pmatrix}, \quad b^{\text{sep}} = \begin{pmatrix} \omega_1/2 \\ \omega_1/2 \\ \omega_2 \end{pmatrix}.$$

For this ordering, the canonical contrast matrix has its sole column

$$N = (1, -1, 0)'.$$

Direct multiplication yields

$$N'\Omega b^{\text{sep}} = -\frac{3\omega_2}{32}, \quad N'\Omega N = \frac{3}{8}.$$

Since $\omega_2 \neq 0$, the diagnostic is nonzero, so this invariant belongs to $\mathcal{E}_+$. Applying the frontier identity once more gives

$$V_{\text{sep}} - V_{\text{joint}} = \frac{(-3\omega_2/32)^2}{3/8} = \frac{3\omega_2^2}{128} > 0.$$

The selected law is the $(\rho, \delta) = (1/2, 1/16)$ member of the open bounded nonzero-effect family in Theorem 4, completing the proof. $\square$

Justification. Turns the equality frontier into a computable invariant and specializes the open family to an explicit compatible witness with variance gap $3\omega_2^2/128$. *Gap.* The partition retains the full compatible covariance map and is not an explanatory classification from an unlabelled graph alone. *Consumer.* A downstream analyst can evaluate $N'\mathfrak{C}(P)b^{\text{sep}}(P)$ to determine constant-span variance loss; the displayed witness proves that the strict class is genuinely nonempty.

Theorem 4 (thm:open-strict-gap-family). *Take $T = 3$, $\mathcal{K} = \{(2, 2), (2, 3)\}$, $\mathcal{J}_{(2,2)} = \{3, \infty\}$, $\mathcal{J}_{(2,3)} = \{\infty\}$, $\varepsilon = 1/8$, $C_4 = 1$, and $\kappa = 1/8$. Fix $\omega = (\omega_1, \omega_2)'$ with $\omega_1 + \omega_2 = 1$ and $\omega_2 \neq 0$. For every*

$$0 < |\rho| < \frac{\sqrt{7}}{3} \quad \text{and} \quad 0 < |\delta| < \frac{1}{8},$$

there is a bounded compatible law $Q_{\rho,\delta} \in \mathcal{M}$, with nonzero constant post-adoption effects equal to δ , for which, in the coordinate order $((2, 2), 3), ((2, 2), \infty), ((2, 3), \infty)$,

$$\Omega(\rho) = \begin{pmatrix} 3/8 & 3/16 & 0 \\ 3/16 & 3/8 & 3\rho/16 \\ 0 & 3\rho/16 & 3/8 \end{pmatrix}, \quad b^{\text{sep}} = \begin{pmatrix} \omega_1/2 \\ \omega_1/2 \\ \omega_2 \end{pmatrix},$$

$$b^{\text{joint}} = \begin{pmatrix} \omega_1/2 + \rho\omega_2/2 \\ \omega_1/2 - \rho\omega_2/2 \\ \omega_2 \end{pmatrix}, \quad V_{\text{sep}} - V_{\text{joint}} = \frac{3\rho^2\omega_2^2}{32} > 0.$$

Thus the smallest architecture contains a nondegenerate open two-parameter family of strict-gap laws, rather than only a zero-effect isolated witness.

Proof. Fix admissible ρ and δ . Let $X_i = x_0$ almost surely, and let (G_i, S_i) be uniform on

$$\{2, 3, \infty\} \times \{0, 1\}.$$

Conditional on $(G_i, S_i) = (a, s)$, generate $(A_i, B_i) \in \{-1, 1\}^2$ according to

$$Q_{\rho, \delta}(A_i = u, B_i = v \mid G_i = a, S_i = s) = \frac{1 + r_a uv}{4}, \quad u, v \in \{-1, 1\},$$

where $r_2 = r_3 = 0$ and $r_\infty = \rho$. This is a probability distribution because $|\rho| < \sqrt{7}/3 < 1$. It satisfies

$$E[A_i \mid G_i = a, S_i = s] = E[B_i \mid G_i = a, S_i = s] = 0,$$

$$E[A_i^2 \mid G_i = a, S_i = s] = E[B_i^2 \mid G_i = a, S_i = s] = 1, \quad E[A_i B_i \mid G_i = a, S_i = s] = r_a.$$

Set

$$Z_{i1} = 0, \quad Z_{i2} = \frac{A_i}{8}, \quad Z_{i3} = \frac{B_i}{8}.$$

For $a \in \{2, 3, \infty\}$, $s \in \{0, 1\}$, and $t \in \{1, 2, 3\}$, define

$$Y_{it}(a, s) = Z_{it} + \delta s \mathbf{1}\{a < \infty\} \mathbf{1}\{t \geq a\}, \quad Y_{it} = Y_{it}(G_i, S_i).$$

These specifications determine the full-data law $Q_{\rho, \delta}$.

Consistency holds by construction. If $t < a$, then $Y_{it}(a, 1) = Z_{it} = Y_{it}(\infty, 1)$, proving no anticipation. For $s = 0$, $Y_{it}(a, 0) = Z_{it} = Y_{it}(\infty, 0)$, proving no spillover to ineligible. Moreover, under never enabling both eligibility states have outcome history (Z_{i1}, Z_{i2}, Z_{i3}) . Hence every untreated eligibility contrast in every retained pairwise conditional DDD restriction is zero, so all such restrictions hold.

The retained cells are the six elements of $\{2, 3, \infty\} \times \{0, 1\}$, so $L_{\text{ret}} = 6$. Since $X_i = x_0$,

$$p_{a,s}(x_0) = \frac{1}{6}, \quad 0 < \varepsilon = \frac{1}{8} < \frac{1}{6} = L_{\text{ret}}^{-1}, \quad p_{a,s}(x_0) \geq \varepsilon.$$

Thus overlap holds. Also

$$|Y_{it}(a, s)| \leq \frac{1}{8} + |\delta| < \frac{1}{4},$$

and therefore $E_P[\max_t |Y_{it}|^4] < 1 = C_4$.

Write $k_1 = (2, 2)$, $k_2 = (2, 3)$, and order the scores as

$$j_1 = (k_1, 3), \quad j_2 = (k_1, \infty), \quad j_3 = (k_2, \infty).$$

The cell regressions are

$$m_{a,s,k_1}(x_0) = \delta s \mathbf{1}\{a = 2\}, \quad m_{a,s,k_2}(x_0) = \delta s \mathbf{1}\{a \in \{2, 3\}\}.$$

Consequently

$$d_{k_1,3}(x_0) = d_{k_1,\infty}(x_0) = d_{k_2,\infty}(x_0) = \delta, \quad \Theta_{k_1}(P) = \Theta_{k_2}(P) = \delta.$$

Thus every target-cell contrast term in the centered scores vanishes. The residuals are

$$\Delta Y_{i,k_1} - m_{a,s,k_1}(x_0) = \frac{A_i}{8}, \quad \Delta Y_{i,k_2} - m_{a,s,k_2}(x_0) = \frac{B_i}{8}.$$

Because $p_{2,1} = \pi_{2,1} = p_{a,s} = 1/6$, every nonzero residual loading is $6\sigma_{a,s}^{2,c}$.

Each score contains four mutually exclusive cells, each with probability $1/6$, squared loading 36 , and residual variance $1/64$. Therefore every diagonal covariance is $3/8$. Scores j_1 and j_2 share exactly the two cohort-2 cells with equal signs, giving covariance $3/16$. Scores j_1 and j_3 share cohort 2, where $E[A_i B_i \mid G_i = 2, S_i] = 0$, giving covariance

zero. Scores j_2 and j_3 have zero covariance on cohort 2 and covariance $\rho/64$ on each of the two never-enabled cells, giving covariance $3\rho/16$. Lemma 2 thus yields

$$\Omega(\rho) = \begin{pmatrix} 3/8 & 3/16 & 0 \\ 3/16 & 3/8 & 3\rho/16 \\ 0 & 3\rho/16 & 3/8 \end{pmatrix} = \frac{3}{16} \begin{pmatrix} 2 & 1 & 0 \\ 1 & 2 & \rho \\ 0 & \rho & 2 \end{pmatrix}.$$

The eigenvalues of the last unscaled matrix are

$$2, \quad 2 - \sqrt{1 + \rho^2}, \quad 2 + \sqrt{1 + \rho^2},$$

because its characteristic polynomial is $(2 - \lambda)\{(2 - \lambda)^2 - (1 + \rho^2)\}$. Since $\rho^2 < 7/9$, $\sqrt{1 + \rho^2} < 4/3$, and hence

$$\lambda_{\min}\{\Omega(\rho)\} > \frac{1}{8}, \quad \lambda_{\max}\{\Omega(\rho)\} < \frac{5}{8} < 8.$$

Therefore $8^{-1}I_3 \preceq \Omega(\rho) \preceq 8I_3$. Every defining condition of $\mathcal{M}$ has now been verified, so $Q_{\rho,\delta} \in \mathcal{M}$.

The first covariance block is $(3/16) \begin{pmatrix} 2 & 1 \\ 1 & 2 \end{pmatrix}$, whose normalized inverse loading is $(1/2, 1/2)'$; the second block is a singleton. Thus

$$b^{\text{sep}} = \begin{pmatrix} \omega_1/2 \\ \omega_1/2 \\ \omega_2 \end{pmatrix}.$$

The joint constraints are $b_1 + b_2 = \omega_1$ and $b_3 = \omega_2$, so every feasible vector is

$$b(q) = \begin{pmatrix} \omega_1/2 + q \\ \omega_1/2 - q \\ \omega_2 \end{pmatrix}.$$

With $h = (1, -1, 0)'$, direct differentiation gives

$$\frac{d}{dq} \{b(q)' \Omega b(q)\} = \frac{3}{8}(2q - \rho\omega_2), \quad \frac{d^2}{dq^2} \{b(q)' \Omega b(q)\} = \frac{3}{4} > 0.$$

The unique minimizer occurs at $q = \rho\omega_2/2$, proving

$$b^{\text{joint}} = \begin{pmatrix} \omega_1/2 + \rho\omega_2/2 \\ \omega_1/2 - \rho\omega_2/2 \\ \omega_2 \end{pmatrix}.$$

The difference is $b^{\text{sep}} - b^{\text{joint}} = -(\rho\omega_2/2)h$, and $h'\Omega h = 3/8$. Orthogonality to the constrained minimizer therefore gives

$$V_{\text{sep}} - V_{\text{joint}} = \frac{\rho^2\omega_2^2}{4} \frac{3}{8} = \frac{3\rho^2\omega_2^2}{32} > 0.$$

Finally, every post-adoption eligible effect equals $\delta \neq 0$. The parameter set $\{(\rho, \delta) : 0 < |\rho| < \sqrt{7}/3, 0 < |\delta| < 1/8\}$ is open and two-dimensional, and the laws vary with both parameters. This proves the nondegenerate open family. $\square$

Justification. Constructs an open bounded compatible family with nonzero effects and a strictly positive constant-span variance gap. *Gap.* With constant X , ρ creates cross-event residual covariance through reuse of the never-enabled cohort; it does not witness covariate-varying relative precision. *Consumer.* Shows that the strict side of the frontier is nonempty in the smallest architecture.

Lemma 6 (lem:retained-cell-count-bounds). *Let*

$$\mathcal{C}_{\text{ret}} = \{(a, s) : s \in \{0, 1\}, \exists k = (g, t) \in \mathcal{K}, c \in \mathcal{J}_k, a \in \{g, c\}\}, \quad L_{\text{ret}} = |\mathcal{C}_{\text{ret}}|.$$

Then $L_{\text{ret}} \geq 4$. Consequently $0 < \varepsilon < L_{\text{ret}}^{-1}$ implies $0 < \varepsilon < 1/4$. Under cell overlap, for every retained score coordinate $((g, t), c)$ and every $(a, s) \in \{g, c\} \times \{0, 1\}$,

$$\varepsilon \leq p_{a,s}(x) \leq 1 - 3\varepsilon \leq 1 - \varepsilon/2$$

for P_X -almost every x . In the four-cell construction of Proposition 1, $L_{\text{ret}} = 4$. In the architecture of Theorem 4, $L_{\text{ret}} = 6$, and its choice $\varepsilon = 1/8$ satisfies $1/8 < 1/6 = L_{\text{ret}}^{-1}$.

Proof. Because $\mathcal{K} \neq \emptyset$ and every $\mathcal{J}_k \neq \emptyset$, choose $k = (g, t) \in \mathcal{K}$ and $c \in \mathcal{J}_k$. By the definition of a retained comparison cohort, $c > t$, while $k \in \mathcal{K}$ gives $t \geq g$. Hence $c \neq g$. It follows that

$$(g, 0), \quad (g, 1), \quad (c, 0), \quad (c, 1)$$

are four distinct members of $\mathcal{C}_{\text{ret}}$. Therefore

$$L_{\text{ret}} = |\mathcal{C}_{\text{ret}}| \geq 4.$$

Since $L_{\text{ret}} > 0$, the stipulated domain $0 < \varepsilon < L_{\text{ret}}^{-1}$ yields

$$0 < \varepsilon < L_{\text{ret}}^{-1} \leq \frac{1}{4},$$

and thus $0 < \varepsilon < 1/4$.

Fix a retained score coordinate $((g, t), c)$. The cell-overlap assumption holds simultaneously outside a P_X -null set for all cells in the finite set $\mathcal{C}_{\text{ret}}$. On that common full-measure set, fix $(a, s) \in \{g, c\} \times \{0, 1\}$. The overlap assumption gives

$$p_{a,s}(x) \geq \varepsilon.$$

The other three members of $\{g, c\} \times \{0, 1\}$ are distinct retained cells, so their conditional probabilities are each at least ε . Conditional cell probabilities are nonnegative and sum to one over all values of (G_i, S_i) . Consequently,

$$\begin{aligned} p_{a,s}(x) &\leq 1 - \sum_{(a', s') \in (\{g, c\} \times \{0, 1\}) \setminus \{(a, s)\}} p_{a', s'}(x) \\ &\leq 1 - 3\varepsilon. \end{aligned}$$

Finally, $\varepsilon > 0$ implies $3\varepsilon \geq \varepsilon/2$, and hence

$$1 - 3\varepsilon \leq 1 - \varepsilon/2.$$

This proves the asserted bounds for P_X -almost every x .

In the construction of Proposition 1, the only retained target cohort is 2 and the only retained comparison cohort is 3. Thus

$$\mathcal{C}_{\text{ret}} = \{2, 3\} \times \{0, 1\}, \quad L_{\text{ret}} = 4.$$

In the architecture of Theorem 4, the retained target cohort is 2, while the retained comparison cohorts are 3 and ∞ . Therefore

$$\mathcal{C}_{\text{ret}} = \{2, 3, \infty\} \times \{0, 1\}, \quad L_{\text{ret}} = 6.$$

For that architecture, $\varepsilon = 1/8$, and direct comparison gives

$$\frac{1}{8} < \frac{1}{6} = L_{\text{ret}}^{-1}.$$

All conclusions follow. □

Justification. Supplies the audited combinatorial and overlap bounds required by the clipped score argument and verifies the exact retained-cell counts in both explicit constructions. *Gap.* This is a supporting finite-cell calculation, not an independent causal or efficiency result. *Consumer.* Downstream proofs may use $L_{\text{ret}} \geq 4$, $p_{a,s} \leq 1 - 3\varepsilon \leq 1 - \varepsilon/2$, and the architecture-specific counts without silently strengthening overlap.

7 Interpretation

The diagnostic $N'\Omega b^{\text{sep}}$ tests whether the covariance-adjusted marginal value of a retained comparison score is constant within each group-event block once DDD shared-cell cross-block covariance is included. It vanishes exactly when the published separate within-block weighting is also optimal for the prespecified aggregate. The labelled invariant is only an evaluable bookkeeping representation of this frontier. The bounded three-coordinate family shows that a strict gain is compatible with feasible overlap, positive-definite covariance, bounded outcomes, and nonzero effects.

Technical internal limitation (diagnostic only)

Optimality is only over constant linear combinations of the fixed retained Ortiz-Villavicencio–Sant’Anna pairwise-score vector. The note does not characterize the tangent space of the full conditional-moment staggered-DDD model and therefore does not provide its efficient influence function or semiparametric efficiency bound; it is not a substitute for Caron’s semiparametric analysis of a distinct subgroup-ATT-difference target. It also does not establish a covariate-varying relative-precision frontier or a rate or test for the descriptive gap diagnostic.

8 Honest scope

The full-data law Q defines the causal target, while the observed-data pushforward $P = Q_O$ defines the scores, nuisance functions, covariance, estimators, and limit law; the identification lemma supplies the bridge. The horizon, retained comparisons, score dimension, and aggregate weights are fixed. Inference is pointwise for a fixed law and supplied learner sequences satisfying foldwise independence, separate $L_4(P_X)$ nuisance consistency, and the product rate. No uniform learner-achievability or uniform-coverage claim is made. Shared-unit covariance accounting and generic constrained-GMM/SUR minimization are treated as prior art; the contribution is restricted to the DDD-specific shared-cell mapping, feasible joint constant-span estimator, exact frontier, diagnostic, and compatible strict-gap family.

References

- [1] Marcelo Ortiz-Villavicencio and Pedro H. C. Sant’Anna. Better Understanding Triple Differences Estimators. arXiv:2505.09942v3, 2025.
- [2] Xiaohong Chen, Pedro H. C. Sant’Anna, and Haitian Xie. Efficient Difference-in-Differences and Event Study Estimators. arXiv:2506.17729v1, 2025.
- [3] Sina Akbari, Negar Kiyavash, and AmirEmad Ghassami. Semiparametric Triple Difference Estimators. arXiv:2502.19788v3, 2025.
- [4] Laura Caron. Triple Difference Designs with Heterogeneous Treatment Effects. arXiv:2502.19620v2, 2025.
- [5] Daniel Akerberg, Xiaohong Chen, Jinyong Hahn, and Zhipeng Liao. Asymptotic Efficiency of Semiparametric Two-Step GMM. Review of Economic Studies, 81(3):919–943, 2014.
- [6] Arnold Zellner. An Efficient Method of Estimating Seemingly Unrelated Regressions and Tests for Aggregation Bias. Journal of the American Statistical Association, 57(298):348–368, 1962.
- [7] Lars Peter Hansen. Large Sample Properties of Generalized Method of Moments Estimators. Econometrica, 50(4):1029–1054, 1982.

- [8] Victor Chernozhukov, Denis Chetverikov, Mert Demirer, Esther Duflo, Christian Hansen, Whitney Newey, and James Robins. Double/Debiased Machine Learning for Treatment and Structural Parameters. *The Econometrics Journal*, 21(1):C1–C68, 2018.
- [9] Andreas Olden and Jarle Moen. The Triple Difference Estimator. *The Econometrics Journal*, 25(3):531–553, 2022.
- [10] Meng Hsuan Hsieh. Stacked Triple Differences. arXiv:2604.22982v1, 2026. Verified source locator: <https://arxiv.org/abs/2604.22982>, Section 6, Theorem 6.3 and Propositions 6.4–6.5.
- [11] A. W. van der Vaart. *Asymptotic Statistics*. Cambridge University Press, 1998. DOI: 10.1017/CBO9780511802256.